# MILITARY SPECIFICATION

## MODULES, STANDARD ELECTRONIC

## DYNAMIC RANDOM ACCESS MEMORY MODULE, KEY CODE CJK

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

### 1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic dynamic Random Access Memory (RAM) module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbook. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

DOD-HDBK-217 - Reliability Prediction of Electronic Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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## 3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 4.4982 failures per million operating hours when predicted in accordance with DOD-HDBK-217E.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be CJK.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 100 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.



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TABLE I. Module electrical requirements.

| Test cond | Parameter 5/ | Ref test table | Test type 1/-2/ | 25° ± 5°C 4/ | | 0° ± 5°C 4/ | | 85° ± 5°C 4/ | | Units |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Min | Max | Min | Max | Min | Max | |
| A1 | Isolation | II | S | 10 | INF | 10 | INF | 10 | INF | MOhm |
| A2-A43 | Isolation | II | X | 10 | INF | 10 | INF | 10 | INF | MOhm |
| B1-B16 | Continuity | III | X | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | Ohm |
| B17-B32 | Continuity | III | Q | 0.0 | 0.1 | 0.0 | 0.1 | 0.0 | 0.1 | Ohm |
| C | Vcc stress | IV | S | Module shall pass all remaining tests | | | | | | |
| D1,D2 | Iih | V | X | -80 | 80 | -80 | 80 | -80 | 80 | uA |
| D3,D4 | Iih | V | X | -50 | 50 | -50 | 50 | -50 | 50 | uA |
| D5-D13 | Iih | V | X | -130 | 130 | -130 | 130 | -130 | 130 | uA |
| D14-D26 | Iih | V | X | -10 | 10 | -10 | 10 | -10 | 10 | uA |
| E1-E13 | Vic | VI | X | -1.0 | -0.4 | -1.0 | -0.4 | -1.0 | -0.4 | V |
| F1,F2 | Iil | VII | X | -80 | 80 | -80 | 80 | -80 | 80 | uA |
| F3,F4 | Iil | VII | X | -50 | 50 | -50 | 50 | -50 | 50 | uA |
| F5-F13 | Iil | VII | X | -130 | 130 | -130 | 130 | -130 | 130 | uA |
| F14-F26 | Iil | VII | X | -10 | 10 | -10 | 10 | -10 | 10 | uA |
| G1-G13 | Voh | VIII | X | 2.4 | 4.5 | 2.4 | 4.5 | 2.4 | 4.5 | V |
| G14 | Voh | VIII | X | 2.4 | 2.6 | 2.4 | 2.6 | 2.4 | 2.8 | V |
| H1-H13 | Vol | IX | X | 0.0 | 0.4 | 0.0 | 0.4 | 0.0 | 0.4 | V |
| J1-J13 | Iozh | X | X | -10 | 10 | -10 | 10 | -10 | 10 | uA |
| K1-K13 | Iozl | XI | X | -10 | 10 | -10 | 10 | -10 | 10 | uA |
| L1 | Icc | XII | X | 39.0 | 700.0 | 39.0 | 700.0 | 39.0 | 700.0 | mA |
| L2 | Icc | XII | X | 6.5 | 130.0 | 6.5 | 130.0 | 6.5 | 130.0 | mA |
| M1,M2 | Functional | XIII | X | See table XIII | | | | | | |
| N1,N2 | Dynamic | XIV | S | See table XIV | | | | | | |
| N3,N4 | Dynamic | XIV | Q | See table XIV | | | | | | |

— See footnotes at end of table





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TABLE I. Module electrical requirements - Continued.

| Test cond | Parameter 5/ | Ref test table | Test type 1/-3/ | 25° ± 5°C 4/ | | 0° ± 5°C 4/ | | 65° ± 5°C 4/ | | Units |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Min | Max | Min | Max | Min | Max | |
| P | PSFC | XVI | S | 14.1 | 26.1 | 14.1 | 26.1 | 14.1 | 26.1 | uF |
| Q | PSTR | XVII | Q | 0 | 400 | 0 | 400 | 0 | 400 | mV |
| R1-R13 | Cin | XVIII | Q | 2 | 20 | 2 | 20 | 2 | 20 | pF |
| R14-R22 | Cin | XVIII | Q | 26 | 195 | 26 | 195 | 26 | 195 | pF |
| R23,R24 | Cin | XVIII | Q | 16 | 120 | 16 | 120 | 16 | 120 | pF |
| R25,R26 | Cin | XVIII | Q | 10 | 75 | 10 | 75 | 10 | 75 | pF |
| S1-S12 | Cout | XIX | Q | 2 | 25 | 2 | 25 | 2 | 25 | pF |
| T1-T13 | Disable time | XX | Q | 0 | 46 | 0 | 46 | 0 | 46 | nS |
| T14-T26 | Disable time | XX | Q | 0 | 65 | 0 | 65 | 0 | 65 | nS |
| U | Threshold | XXI | X | See table XXI | | | | | | |

- See footnotes on next page.



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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.



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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/361. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.



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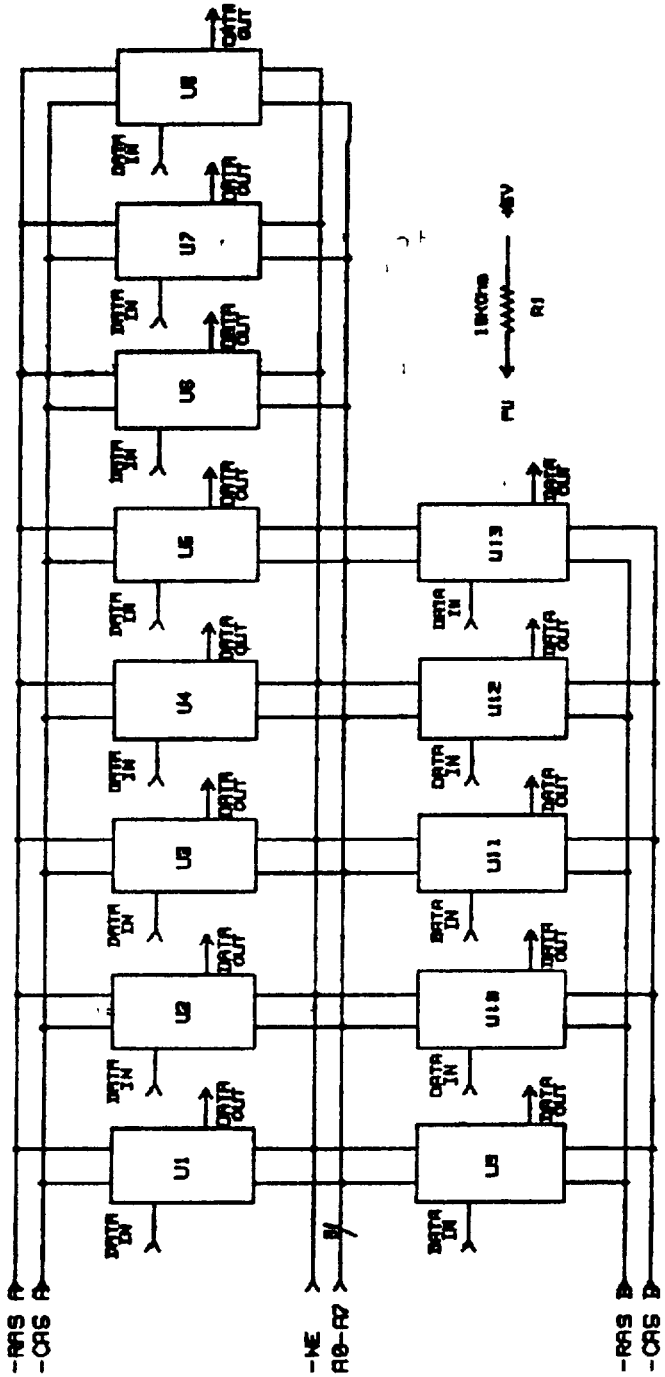



FIGURE 1. CJK functional block diagram.



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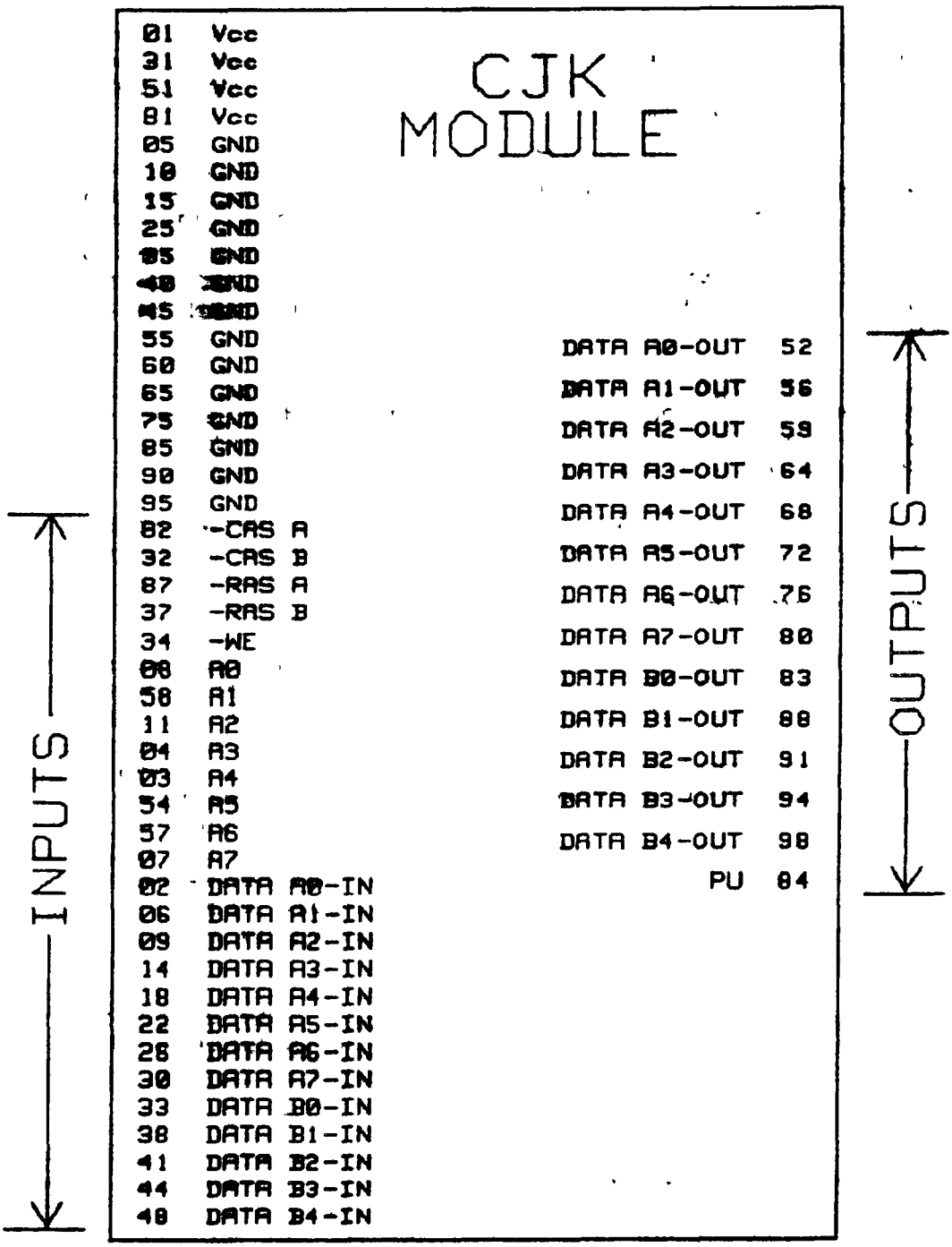



FIGURE 2. CJK pin assignment.



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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. Test conditions A and C are stress tests and shall be performed only at 25 ± 5°C. If test conditions A or C are part of a sequence of tests, the order of testing shall be as follows: A and C where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

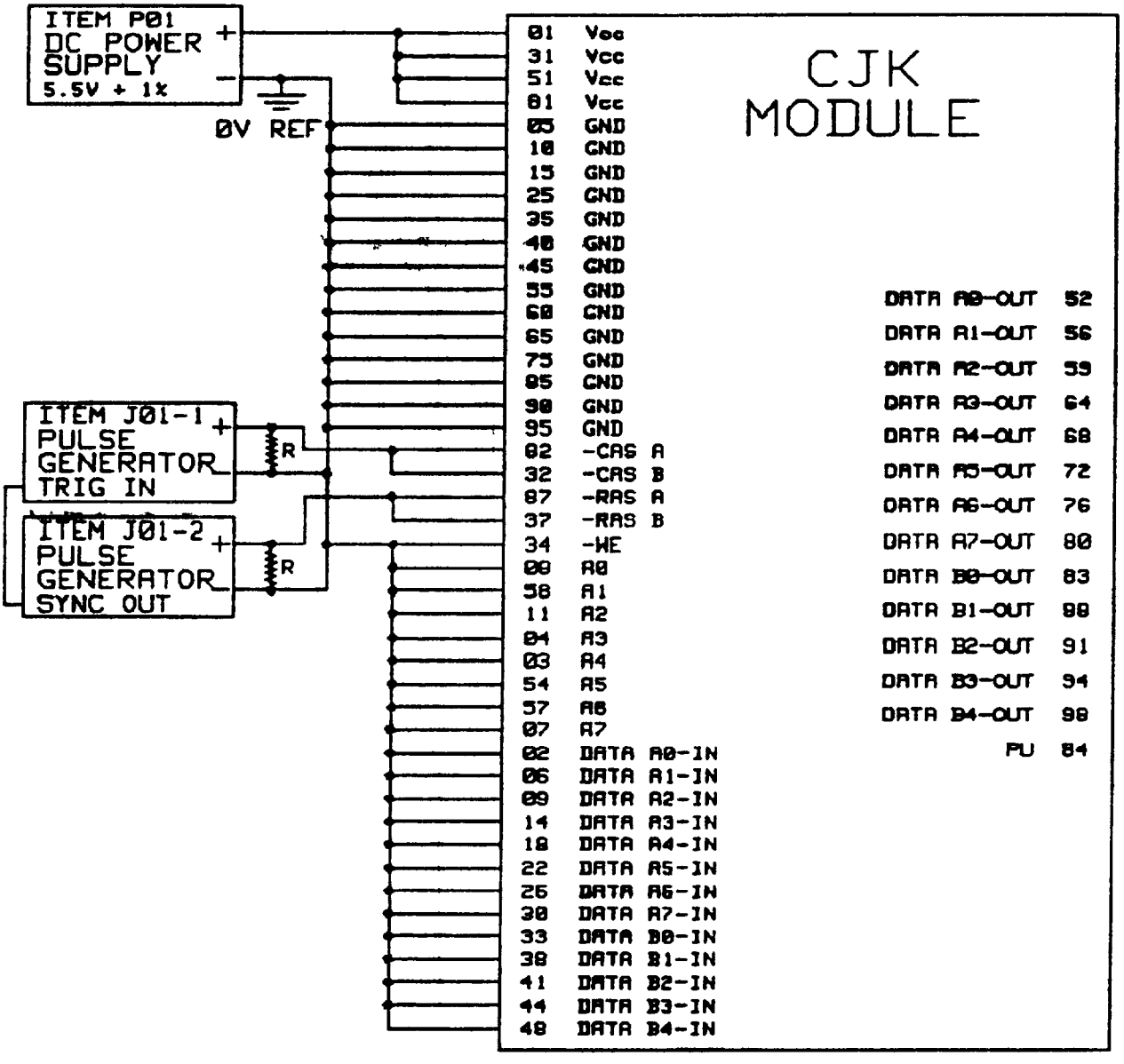
4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, D through Q, T and U shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3. The input waveforms are shown on figure 4.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 3 and 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition P, shall be performed using the initial limits at 25 ± 5°C ambient.



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NOTES:

1. The load is as specified in the output voltage test.
2. R = 51 Ohm ± 5 percent, 1 W.

FIGURE 3. CJK operating temperature, component temperature and life test circuit.



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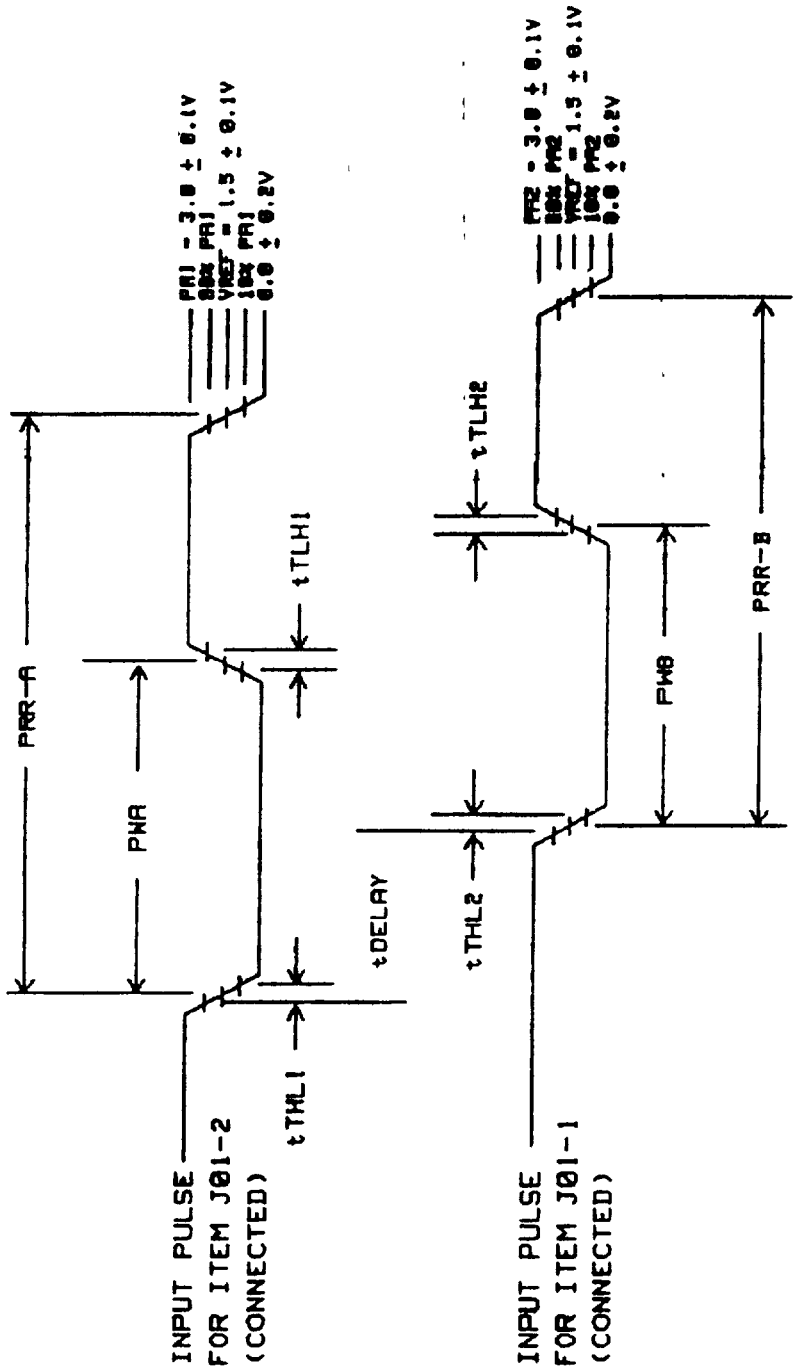



FIGURE 4. CJK operating temperature, component temperature, and life test circuit waveforms.



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NOTES:

1. The input parameters are as follows:

   PWA = 200 ± 20 nS.
   tTLH1 = tTHL1 = 3.5 ± 1.5 nS.
   tDELAY = 35 ± 5 nS.
   PWB = 200 ± 20 nS.
   tTLH2 = tTHL2 = 3.5 ± 1.5 nS.
   PRR-A = PRR-B = 350 ± 35 nS.

2. All voltages are referenced to ground.



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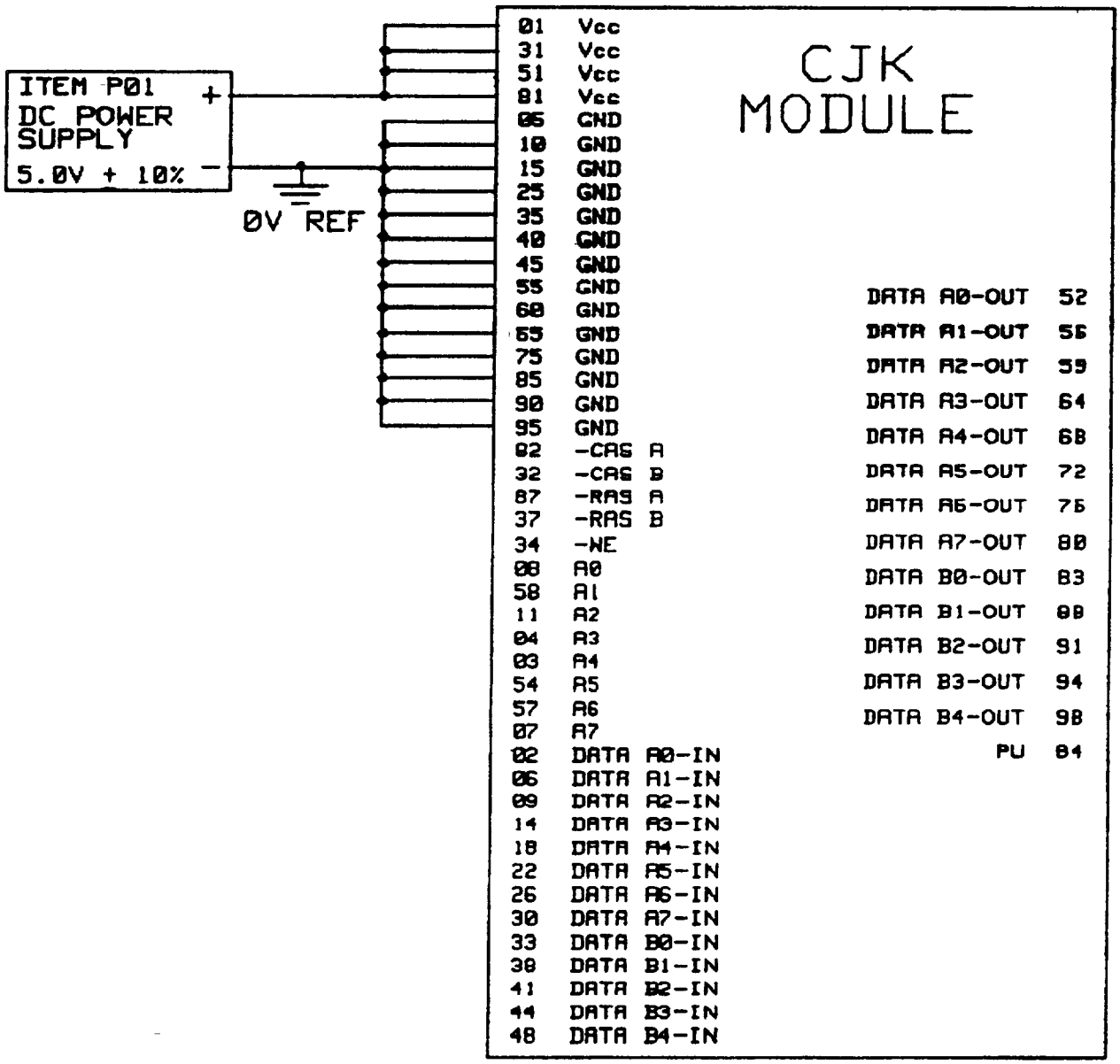



FIGURE 5. CJK vibration test circuit.



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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 3 and 4. Following the life test and while still at the maximum class temperature, electrical tests B, D through Q, T and U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests A1, B, D through Q, T and U shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.



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## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Ibd = input breakdown current.
(d) Icc = power supply current.
(e) Iih = input high level current.
(f) Iil = input low level current.
(g) Iin = input forcing current.
(h) Ioh = output high level current.
(i) Iol = output low level current.
(j) Ios = output short circuit current.
(k) Iozh = output high level leakage current.
(l) Iozl = output low level leakage current.
(m) NA = not applicable.
(n) PSFC = power supply filter capacitance.
(o) PSTA = power supply transient amplitude.
(p) Vic = input clamp diode voltage.
(q) Vih = input high level voltage.
(r) Vil = input low level voltage.
(s) Vin = input forcing voltage.
(t) Voh = output high level voltage.
(u) Vol = output low level voltage.
(v) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 4.293 W (max); 2.989 W (typical).



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6.3.1 Input parameters.

| Signal | Vic min (V) | Vil max (V) | Vih min (V) | Iil max (uA) | Iih max (uA) | Cin max (pF) |
|---|---|---|---|---|---|---|
| A0 | NA | 0.8 | 2.4 | 130 | 130 | 195 |
| A1 | NA | 0.8 | 2.4 | 130 | 130 | 195 |
| A2 | NA | 0.8 | 2.4 | 130 | 130 | 195 |
| A3 | NA | 0.8 | 2.4 | 130 | 130 | 195 |
| A4 | NA | 0.8 | 2.4 | 130 | 130 | 195 |
| A5 | NA | 0.8 | 2.4 | 130 | 130 | 195 |
| A6 | NA | 0.8 | 2.4 | 130 | 130 | 195 |
| A7 | NA | 0.8 | 2.4 | 130 | 130 | 195 |
| -WE | NA | 0.8 | 2.4 | 130 | 130 | 195 |
| -RAS A | NA | 0.8 | 2.4 | 80 | 80 | 120 |
| -CAS A | NA | 0.8 | 2.4 | 80 | 80 | 120 |
| -RAS B | NA | 0.8 | 2.4 | 50 | 50 | 75 |
| -CAS B | NA | 0.8 | 2.4 | 50 | 50 | 75 |
| DATA A0-IN | -1.0 | 0.8 | 2.4 | 10 | 10 | 20 |



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6.3.1 Input parameters - Continued.

| Signal | Vic min (V) | Vil max (V) | Vih min (V) | Iil max (uA) | Iih max (uA) | Cin max (pF) |
|---|---|---|---|---|---|---|
| DATA A1-IN | -1.0 | 0.8 | 2.4 | 10 | 10 | 20 |
| DATA A2-IN | -1.0 | 0.8 | 2.4 | 10 | 10 | 20 |
| DATA A3-IN | -1.0 | 0.8 | 2.4 | 10 | 10 | 20 |
| DATA A4-IN | -1.0 | 0.8 | 2.4 | 10 | 10 | 20 |
| DATA A5-IN | -1.0 | 0.8 | 2.4 | 10 | 10 | 20 |
| DATA A6-IN | -1.0 | 0.8 | 2.4 | 10 | 10 | 20 |
| DATA A7-IN | -1.0 | 0.8 | 2.4 | 10 | 10 | 20 |
| DATA B0-IN | -1.0 | 0.8 | 2.4 | 10 | 10 | 20 |
| DATA B1-IN | -1.0 | 0.8 | 2.4 | 10 | 10 | 20 |
| DATA B2-IN | -1.0 | 0.8 | 2.4 | 10 | 10 | 20 |
| DATA B3-IN | -1.0 | 0.8 | 2.4 | 10 | 10 | 20 |
| DATA B4-IN | -1.0 | 0.8 | 2.4 | 10 | 10 | 20 |

6.3.1.1 Input protection. Schottky clamp diodes are supplied on DATA A0-IN through DATA A7-IN and DATA B0-IN through DATA B4-IN to protect against undershoot. Proper termination techniques, such as series damping resistors, should be used at the driver for A0-A7, -RAS A, -RAS B, -CAS A, -CAS B and -WE to limit undershoot at the module inputs to -1.0 volt.



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6.3.2 Output parameters.

| Signal | Voh min (V) | Vol max (V) | Ioh max (mA) | Iol max (mA) | Iozh max (uA) | Iozl max (uA) | Cout max (pF) |
|---|---|---|---|---|---|---|---|
| DATA A0-OUT | 2.4 | 0.4 | -5.0 | 4.0 | 10 | 10 | 25 |
| DATA A1-OUT | 2.4 | 0.4 | -5.0 | 4.0 | 10 | 10 | 25 |
| DATA A2-OUT | 2.4 | 0.4 | -5.0 | 4.0 | 10 | 10 | 25 |
| DATA A3-OUT | 2.4 | 0.4 | -5.0 | 4.0 | 10 | 10 | 25 |
| DATA A4-OUT | 2.4 | 0.4 | -5.0 | 4.0 | 10 | 10 | 25 |
| DATA A5-OUT | 2.4 | 0.4 | -5.0 | 4.0 | 10 | 10 | 25 |
| DATA A6-OUT | 2.4 | 0.4 | -5.0 | 4.0 | 10 | 10 | 25 |
| DATA A7-OUT | 2.4 | 0.4 | -5.0 | 4.0 | 10 | 10 | 25 |
| DATA B0-OUT | 2.4 | 0.4 | -5.0 | 4.0 | 10 | 10 | 25 |
| DATA B1-OUT | 2.4 | 0.4 | -5.0 | 4.0 | 10 | 10 | 25 |
| DATA B2-OUT | 2.4 | 0.4 | -5.0 | 4.0 | 10 | 10 | 25 |
| DATA B3-OUT | 2.4 | 0.4 | -5.0 | 4.0 | 10 | 10 | 25 |
| DATA B4-OUT | 2.4 | 0.4 | -5.0 | 4.0 | 10 | 10 | 25 |
| PU | 2.4 | NA | 0.2 | NA | NA | NA | NA |



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6.3.3 Dynamic parameters. The dynamic parameters appear with the dynamic test.

6.4 Module function. This module contains a 13 bit by 64K Dynamic RAM (DRAM) memory which is partitioned into an 8-bit DRAM and 5-bit DRAM group. The 8-bit DRAM group is for data bit storage and the 5-bit DRAM group is normally used for check bit storage when used with an error correction circuit. A pullup resistor (10 Kohm) is supplied so the 8-bit user can "power down" the 5-bit group to minimize power dissipation. Schottky clamp diodes are supplied on DATA A0-IN - DATA A7-IN and DATA B0-IN - DATA B4-IN to protect against undershoot. Proper termination techniques, such as series damping resistors, should be used at the driver for A0-A7, -RAS A, -RAS B, -CAS A, -CAS B, and -WE to limit undershoot at the DRAM inputs to -1.0 volt. The page mode operation allows for successive memory operations at multiple column locations within the same row address. This is done by strobing the row address into the memory and maintaining the -RAS A, -RAS B signals low (active) throughout all successive memory cycles in which the row address is common. By using page mode, effective memory cycle times are reduced. Refresh is accomplished by performing a memory cycle at each of the 256 row addresses within each 4 mS interval. Although any normal memory cycle will perform the required refreshing, this function is most easily accomplished with "RAS-only" cycles. Note that all 8 address lines (A0-A7) are used for refresh. A complete refresh interval is accomplished by sequencing the address lines from a count of 0 through 255. The RAS-only refresh cycle requires that a refresh address be valid at the address inputs when -RAS A and -RAS B go low (active). The state of the data output during a RAS-only refresh is controlled by -CAS A and -CAS B. If -CAS A and -CAS B are high (inactive) during the entire time that -RAS A and -RAS B are low, the ouput will remain in the high impedance state. If -CAS A and -CAS B are low (active), the entire time that -RAS A and -RAS B are asserted (low), the output will remain in the same state that it was prior to the issuance of the -RAS A and -RAS B signal. If -CAS A and -CAS B make a low to high transition during the RAS-only refresh cycle, the output will assume the high impedance state. However, -CAS A and -CAS B may not make a high to low transition during the RAS-only refresh cycle since the device interprets this as a normal read or write type cycle. A RAS-only refresh cycle may take place while maintaining valid output data by extending the -CAS A and -CAS B active time from a previous memory read cycle. This feature is referred to as a hidden refresh. This also allows the output data from the previous memory cycle to be held valid for the additional time that -CAS A and -CAS B are extended.



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6.4.1 Functional description of I/O pins.

| Signal name | Pin number | I/O | Functional description |
|---|---|---|---|
| -RAS A | 87 | I | Row Address Strobe used to transfer in the row address being selected. |
| -RAS B | 37 | I | Row Address Strobe used to transfer in the row address being selected. |
| -CAS A | 82 | I | Column Address Strobe used to transfer in the column address being selected. |
| -CAS B | 32 | I | Column Address Strobe used to transfer in the column address being selected. |
| -WE | 34 | I | Write Enable signal used to write data into the module when low. |
| A0 | 8 | I | Address bus used to select memory locations. |
| A1 | 58 | I | |
| A2 | 11 | I | |
| A3 | 4 | I | |
| A4 | 3 | I | |
| A5 | 54 | I | |
| A6 | 57 | I | |
| A7 | 7 | I | |



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6.4.1 Functional description of I/O pins - Continued.

| Signal name | Pin number | I/O | Functional description |
|---|---|---|---|
| DATA A0-IN | 2 | I | Data inputs partitioned into an 8-bit and 5-bit data bus. |
| DATA A1-IN | 6 | I | |
| DATA A2-IN | 9 | I | |
| DATA A3-IN | 14 | I | |
| DATA A4-IN | 18 | I | |
| DATA A5-IN | 22 | I | |
| DATA A6-IN | 26 | I | |
| DATA A7-IN | 30 | I | |
| DATA B0-IN | 33 | I | |
| DATA B1-IN | 38 | I | |
| DATA B2-IN | 41 | I | |
| DATA B3-IN | 44 | I | |
| DATA B4-IN | 48 | I | |
| DATA A0-OUT | 52 | O | Memory data output bus sectioned into an 8-bit and a 5-bit group. |
| DATA A1-OUT | 56 | O | |
| DATA A2-OUT | 59 | O | |
| DATA A3-OUT | 64 | O | |
| DATA A4-OUT | 68 | O | |
| DATA A5-OUT | 72 | O | |
| DATA A6-OUT | 76 | O | |
| DATA A7-OUT | 80 | O | |
| DATA B0-OUT | 83 | O | |
| DATA B1-OUT | 88 | O | |
| DATA B2-OUT | 91 | O | |
| DATA B3-OUT | 94 | O | |
| DATA B4-OUT | 98 | O | |
| PU | 84 | O | General purpose pullup resistor (10 Kohm) that can be used to "power down" 5-bit group to minimize power dissipation. |



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6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II.

(b) Continuity test - table III.

(c) Vcc stress test - table IV.

(d) Iih test - table V.

(e) Vic test - table VI.

(f) Iil test - table VII.

(g) Voh test - table VIII, figure 6.

(h) Vol test - table IX, figure 6.

(i) Iozh test - table X.

(j) Iozl test - table XI.

(k) Icc test - table XII.

(l) Functional test - table XIII, figure 7.

(m) Dynamic test - tables XIV and XV, figures 8 and 9.

(n) PSFC test - table XVI.

(o) PSTA test - table XVII, figure 10.

(p) Cin test - table XVIII.

(q) Cout test - table XIX.

(r) Disable time test - table XX, figures 11 and 12.

(s) Threshold test - table XXI.



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6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N738)

Agent: NW



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TABLE II. Isolation test.

| Test cond | Test description | Connect A to pin | Connect B to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| A1 | Isolation, frame | All pins | Frame | 10 | INF | MOhm | 1-7 |
| A2 | Isolation, used to unused pins | All used pins | All unused pins | | | | |
| A3 | Isolation, unused to unused pins | 12 | 13,16,17,19, 20,21,23,24, 27,28,29,36, 39,42,43,46, 47,49,50,53, 61,62,63,66, 67,69,70,71, 73,74,77,78, 79,86,89,92, 93,96,97,99, 100 | | | | |
| A4 | Isolation, unused to unused pins | 13 | 16,17,19,20, 21,23,24,27, 28,29,36,39, 42,43,46,47, 49,50,53,61, 62,63,66,67, 69,70,71,73, 74,77,78,79, 86,89,92,93, 96,97,99,100 | | | | |
| A5 | Isolation, unused to unused pins | 16 | 17,19,20,21, 23,24,27,28, 29,36,39,42, 43,46,47,49, 50,53,61,62, 63,66,67,69, 70,71,73,74, 77,78,79,86, 89,92,93,96, 97,99,100 | 10 | INF | MOhm | 1-7 |



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TABLE II. Isolation test - Continued.

| Test cond | Test description | Connect A to pin | Connect B to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| A6 | Isolation, unused to unused pins | 17 | 19,20,21,23, 24,27,28,29, 36,39,42,43, 46,47,49,50, 53,61,62,63, 66,67,69,70, 71,73,74,77, 78,79,86,89, 92,93,96,97, 99,100 | 10 | INF | MOhm | 1-7 |
| A7 | Isolation, unused to unused pins | 19 | 20,21,23,24, 27,28,29,36, 39,42,43,46, 47,49,50,53, 61,62,63,66, 67,69,70,71, 73,74,77,78, 79,86,89,92, 93,96,97,99, 100 | | | | |
| A8 | Isolation, unused to unused pins | 20 | 21,23,24,27, 28,29,36,39, 42,43,46,47, 49,50,53,61, 62,63,66,67, 69,70,71,73, 74,77,78,79, 86,89,92,93, 96,97,99,100 | 10 | INF | MOhm | 1-7 |



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TABLE II. Isolation test - Continued.

| Test cond | Test description | Connect A to pin | Connect B to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| A9 | Isolation, unused to unused pins | 21 | 23,24,27,28, 29,36,39,42, 43,46,47,49, 50,53,61,62, 63,66,67,69, 70,71,73,74, 77,78,79,86, 89,92,93,96, 97,99,100 | 10 | INF | MOhm | 1-7 |
| A10 | Isolation, unused to unused pins | 23 | 24,27,28,29, 36,39,42,43, 46,47,49,50, 53,61,62,63, 66,67,69,70, 71,73,74,77, 78,79,86,89, 92,93,96,97, 99,100 | | | | |
| A11 | Isolation, unused to unused pins | 24 | 27,28,29,36, 39,42,43,46, 47,49,50,53, 61,62,63,66, 67,69,70,71, 73,74,77,78, 79,86,89,92, 93,96,97,99, 100 | | | | |
| A12 | Isolation, unused to unused pins | 27 | 28,29,36,39, 42,43,46,47, 49,50,53,61, 62,63,66,67, 69,70,71,73, 74,77,78,79, 86,89,92,93, 96,97,99,100 | 10 | INF | MOhm | 1-7 |



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TABLE II. Isolation test - Continued.

| Test cond | Test description | Connect A to pin | Connect B to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| A13 | Isolation, unused to unused pins | 28 | 29,36,39,42, 43,46,47,49, 50,53,61,62, 63,66,67,69, 70,71,73,74, 77,78,79,86, 89,92,93,96, 97,99,100 | 10 | INF | MOhm | 1-7 |
| A14 | Isolation, unused to unused pins | 29 | 36,39,42,43, 46,47,49,50, 53,61,62,63, 66,67,69,70, 71,73,74,77, 78,79,86,89, 92,93,96,97, 99,100 | | | | |
| A15 | Isolation, unused to unused pins | 36 | 39,42,43,46, 47,49,50,53, 61,62,63,66, 67,69,70,71, 73,74,77,78, 79,86,89,92, 93,96,97,99, 100 | | | | |
| A16 | Isolation, unused to unused pins | 39 | 42,43,46,47, 49,50,53,61, 62,63,66,67, 69,70,71,73, 74,77,78,79, 86,89,92,93, 96,97,99,100 | 10 | INF | MOhm | 1-7 |



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TABLE II. Isolation test - Continued.

| Test cond | Test description | Connect A to pin | Connect B to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| A17 | Isolation, unused to unused pins | 42 | 43,46,47,49, 50,53,61,62, 63,66,67,69, 70,71,73,74, 77,78,79,86, 89,92,93,96, 97,99,100 | 10 | INF | MOhm | 1-7 |
| A18 | Isolation, unused to unused pins | 43 | 46,47,49,50, 53,61,62,63, 66,67,69,70, 71,73,74,77, 78,79,86,89, 92,93,96,97, 99,100 | | | | |
| A19 | Isolation, unused to unused pins | 46 | 47,49,50,53, 61,62,63,66, 67,69,70,71, 73,74,77,78, 79,86,89,92, 93,96,97,99, 100 | | | | |
| A20 | Isolation, unused to unused pins | 47 | 49,50,53,61, 62,63,66,67, 69,70,71,73, 74,77,78,79, 86,89,92,93, 96,97,99,100 | | | | |
| A21 | Isolation, unused to unused pins | 49 | 50,53,61,62, 63,66,67,69, 70,71,73,74, 77,78,79,86, 89,92,93,96, 97,99,100 | 10 | INF | MOhm | 1-7 |



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TABLE II. Isolation test - Continued.

| Test cond | Test description | Connect A to pin | Connect B to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| A22 | Isolation, unused to unused pins | 50 | 53,61,62,63, 66,67,69,70, 71,73,74,77, 78,79,86,89, 92,93,96,97, 99,100 | 10 | INF | MOhm | 1-7 |
| A23 | Isolation, unused to unused pins | 53 | 61,62,63,66, 67,69,70,71, 73,74,77,78, 79,86,89,92, 93,96,97,99, 100 | | | | |
| A24 | Isolation, unused to unused pins | 61 | 62,63,66,67, 69,70,71,73, 74,77,78,79, 86,89,92,93, 96,97,99,100 | | | | |
| A25 | Isolation, unused to unused pins | 62 | 63,66,67,69, 70,71,73,74, 77,78,79,86, 89,92,93,96, 97,99,100 | | | | |
| A26 | Isolation, unused to unused pins | 63 | 66,67,69,70, 71,73,74,77, 78,79,86,89, 92,93,96,97, 99,100 | | | | |
| A27 | Isolation, unused to unused pins | 66 | 67,69,70,71, 73,74,77,78, 79,86,89,92, 93,96,97,99, 100 | 10 | INF | MOhm | 1-7 |



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TABLE II. Isolation test - Continued.

| Test cond | Test description | Connect A to pin | Connect B to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| A28 | Isolation, unused to unused pins | 67 | 69,70,71,73, 74,77,78,79, 86,89,92,93, 96,97,99,100 | 10 | INF | MOhm | 1-7 |
| A29 | Isolation, unused to unused pins | 69 | 70,71,73,74, 77,78,79,86, 89,92,93,96, 97,99,100 | | | | |
| A30 | Isolation, unused to unused pins | 70 | 71,73,74,77, 78,79,86,89, 92,93,96,97, 99,100 | | | | |
| A31 | Isolation, unused to unused pins | 71 | 73,74,77,78, 79,86,89,92, 93,96,97,99, 100 | | | | |
| A32 | Isolation, unused to unused pins | 73 | 74,77,78,79, 86,89,92,93, 96,97,99,100 | | | | |
| A33 | Isolation, unused to unused pins | 74 | 77,78,79,86, 89,92,93,96, 97,99,100 | | | | |
| A34 | Isolation, unused to unused pins | 77 | 78,79,86,89, 92,93,96,97, 99,100 | | | | |
| A35 | Isolation, unused to unused pins | 78 | 79,86,89,92, 93,96,97,99, 100 | | | | |
| A36 | Isolation, unused to unused pins | 79 | 86,89,92,93, 96,97,99,100 | 10 | INF | MOhm | 1-7 |



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TABLE II. Isolation test - Continued.

| Test cond | Test description | Connect A to pin | Connect B to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| A37 | Isolation, unused to unused pins | 86 | 89,92,93,96, 97,99,100 | 10 | INF | MOhm | 1-7 |
| A38 | Isolation, unused to unused pins | 89 | 92,93,96,97, 99,100 | | | | |
| A39 | Isolation, unused to unused pins | 92 | 93,96,97,99, 100 | | | | |
| A40 | Isolation, unused to unused pins | 93 | 96,97,99,100 | | | | |
| A41 | Isolation, unused to unused pins | 96 | 97,99,100 | | | | |
| A42 | Isolation, unused to unused pins | 97 | 99,100 | | | | |
| A43 | Isolation, unused to unused pins | 99 | 100 | 10 | INF | MOhm | 1-7 |



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NOTES:

1. Isolation test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Megohm bridge voltage | 100 V<br>30 V | ± 10 percent<br>± 10 percent | A1<br>A2-A43 |
| Isolation measurement (see note 6) | NA<br>NA | ± 500 KOhm<br>± 50 KOhm | A1-A43<br>A1-A43 |

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.



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TABLE III. Continuity test.

| Test cond | Test description | Connect A to pin | Connect B to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| B1 | Continuity, Vcc | 1 | 31 | 0.0 | 1.0 | Ohm | 1-4 |
| B2 | Continuity, Voc | 31 | 51 | | | | |
| B3 | Continuity, Vcc | 51 | 81 | | | | |
| B4 | Continuity, GND | 5 | 10 | | | | |
| B5 | Continuity, GND | 10 | 15 | | | | |
| B6 | Continuity, GND | 15 | 25 | | | | |
| B7 | Continuity, GND | 25 | 35 | | | | |
| B8 | Continuity, GND | 35 | 40 | | | | |
| B9 | Continuity, GND | 40 | 45 | | | | |
| B10 | Continuity, GND | 45 | 55 | | | | |
| B11 | Continuity, GND | 55 | 60 | | | | |
| B12 | Continuity, GND | 60 | 65 | | | | |
| B13 | Continuity, GND | 65 | 75 | | | | |
| B14 | Continuity, GND | 75 | 85 | | | | |
| B15 | Continuity, GND | 85 | 90 | | | | |
| B16 | Continuity, GND | 90 | 95 | 0.0 | 1.0 | Ohm | 1-4 |
| B17 | Continuity, Vcc | 1 | 31 | 0.0 | 0.1 | Ohm | 1-4 |
| B18 | Continuity, Vcc | 31 | 51 | | | | |
| B19 | Continuity, Vcc | 51 | 81 | | | | |
| B20 | Continuity, GND | 5 | 10 | 0.0 | 0.1 | Ohm | 1-4 |



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TABLE III. Continuity test - Continued.

| Test cond | Test description | Connect A to pin | Connect B to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| B21 | Continuity, GND | 10 | 15 | 0.0 | 0.1 | Ohm | 1-4 |
| B22 | Continuity, GND | 15 | 25 | | | | |
| B23 | Continuity, GND | 25 | 35 | | | | |
| B24 | Continuity, GND | 35 | 40 | | | | |
| B25 | Continuity, GND | 40 | 45 | | | | |
| B26 | Continuity, GND | 45 | 55 | | | | |
| B27 | Continuity, GND | 55 | 60 | | | | |
| B28 | Continuity, GND | 60 | 65 | | | | |
| B29 | Continuity, GND | 65 | 75 | | | | |
| B30 | Continuity, GND | 75 | 85 | | | | |
| B31 | Continuity, GND | 85 | 90 | | | | |
| B32 | Continuity, GND | 90 | 95 | 0.0 | 0.1 | Ohm | 1-4 |



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NOTES:

1. Continuity test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Continuity measurement | NA<br>NA | ± 50 mOhm<br>± 5 mOhm | B1-B16<br>B17-B32 |

2. The ohmmeter used shall be item R10.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. Abbreviations used are defined in 6.1.



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TABLE IV. Power supply stress test.

| Test cond | Test description | Vin (V) | Connect Positive lead to | Connect Negative lead to | 25 ± 5°C Initial limits | Notes |
|---|---|---|---|---|---|---|
| C | Power supply stress test | 7.0 | 1,31 51,81 | 10,15,25, 35,40,45, 55,60,65, 75,85,90, 95,5 | Module shall pass all remaining tests. | 1-5 |

NOTES:

1. Power supply stress test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 7.0 V | +0 mV,-70 mV | C |
| Power supply current limit | 1.17 A | NA | C |

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. Abbreviations used are defined in 6.1.



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TABLE V. Input current, high level, test.

| Test cond | Signal name | Pin no. | Vih (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| D1 | -RAS A | 87 | 5.5 | -80 | 80 | uA | 1-3 |
| D2 | -CAS A | 82 | 5.5 | -80 | 80 | uA | 1-3 |
| D3 | -RAS B | 37 | 5.5 | -50 | 50 | uA | 1-3 |
| D4 | -CAS B | 32 | 5.5 | -50 | 50 | uA | 1-3 |
| D5 | -WE | 34 | 5.5 | -130 | 130 | uA | 1-3 |
| D6 | A0 | 8 | | | | | |
| D7 | A1 | 58 | | | | | |
| D8 | A2 | 11 | | | | | |
| D9 | A3 | 4 | | | | | |
| D10 | A4 | 3 | | | | | |
| D11 | A5 | 54 | | | | | |
| D12 | A6 | 57 | | | | | |
| D13 | A7 | 7 | 5.5 | -130 | 130 | uA | 1-3 |
| D14 | DATA A0-IN | 2 | 5.5 | -10 | 10 | uA | 1-3 |
| D15 | DATA A1-IN | 6 | | | | | |
| D16 | DATA A2-IN | 9 | | | | | |
| D17 | DATA A3-IN | 14 | | | | | |
| D18 | DATA A4-IN | 18 | | | | | |
| D19 | DATA A5-IN | 22 | | | | | |
| D20 | DATA A6-IN | 26 | 5.5 | -10 | 10 | uA | 1-3 |



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TABLE V. Input current, high level, test - Continued.

| Test cond | Signal name | Pin no. | Vih (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| D21 | DATA A7-IN | 30 | 5.5 | -10 | 10 | uA | 1-3 |
| D22 | DATA B0-IN | 33 | | | | | |
| D23 | DATA B1-IN | 38 | | | | | |
| D24 | DATA B2-IN | 41 | | | | | |
| D25 | DATA B3-IN | 44 | | | | | |
| D26 | DATA B4-IN | 48 | 5.5 | -10 | 10 | uA | 1-3 |

NOTES:

1. Input current, high level, test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 5.5 V | ± 55 mV | D1-D26 |
| Power supply current limit | 1.17 A | NA | D1-D26 |
| Forcing voltage, (Vin) | 5.5 V | ± 55 mV | D1-D26 |
| Pattern voltage high, (VH) | 3.0 V | ± 100 mV | D1-D26 |
| low, (VL) | 0.0 V | ± 100 mV | D1-D26 |



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NOTES: - Continued.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Input current measurement | NA | ± 8 uA | D1,D2 |
| | NA | ± 5 uA | D3,D4 |
| | NA | ± 13 uA | D5-D13 |
| | NA | ± 1 uA | D14-D26 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.



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TABLE VI. Input clamp diode voltage test.

| Test cond | Signal name | Pin no. | Iin (mA) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| E1 | DATA A0-IN | 2 | -18.0 | -1.0 | -0.4 | V | 1-3 |
| E2 | DATA A1-IN | 6 | | | | | |
| E3 | DATA A2-IN | 9 | | | | | |
| E4 | DATA A3-IN | 14 | | | | | |
| E5 | DATA A4-IN | 18 | | | | | |
| E6 | DATA A5-IN | 22 | | | | | |
| E7 | DATA A6-IN | 26 | | | | | |
| E8 | DATA A7-IN | 30 | | | | | |
| E9 | DATA B0-IN | 33 | | | | | |
| E10 | DATA B1-IN | 38 | | | | | |
| E11 | DATA B2-IN | 41 | | | | | |
| E12 | DATA B3-IN | 44 | | | | | |
| E13 | DATA B4-IN | 48 | -18.0 | -1.0 | -0.4 | V | 1-3 |



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NOTES:

1. Input clamp diode voltage test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 4.5 V | ± 45 mV | E1-E13 |
| Power supply current limit | 1.17 A | NA | E1-E13 |
| Forcing current, (Iin) | -18 mA | ± 1.8 mA | E1-E13 |
| Pattern voltage<br>high, (VH)<br>low, (VL) | <br>3.0 V<br>0.0 V | <br>± 100 mV<br>± 100 mV | <br>E1-E13<br>E1-E13 |
| Input voltage measurement | NA | ± 30 mV | E1-E13 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.



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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VI and the voltage then measured.

3. Abbreviations used are defined in 6.1.



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TABLE VII. Input current, low level, test.

| Test cond | Signal name | Pin no. | Vil (V) | 25 ± 5°C Initial limits Min | Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| F1 | -RAS A | 87 | 0.0 | -80 | 80 | uA | 1-3 |
| F2 | -CAS A | 82 | 0.0 | -80 | 80 | uA | 1-3 |
| F3 | -RAS B | 37 | 0.0 | -50 | 50 | uA | 1-3 |
| F4 | -CAS B | 32 | 0.0 | -50 | 50 | uA | 1-3 |
| F5 | -WE | 34 | 0.0 | -130 | 130 | uA | 1-3 |
| F6 | A0 | 8 | | | | | |
| F7 | A1 | 58 | | | | | |
| F8 | A2 | 11 | | | | | |
| F9 | A3 | 4 | | | | | |
| F10 | A4 | 3 | | | | | |
| F11 | A5 | 54 | | | | | |
| F12 | A6 | 57 | | | | | |
| F13 | A7 | 7 | 0.0 | -130 | 130 | uA | 1-3 |
| F14 | DATA A0-IN | 2 | 0.0 | -10 | 10 | uA | 1-3 |
| F15 | DATA A1-IN | 6 | | | | | |
| F16 | DATA A2-IN | 9 | | | | | |
| F17 | DATA A3-IN | 14 | | | | | |
| F18 | DATA A4-IN | 18 | | | | | |
| F19 | DATA A5-IN | 22 | | | | | |
| F20 | DATA A6-IN | 26 | 0.0 | -10 | 10 | uA | 1-3 |



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TABLE VII. Input current, low level, test - Continued.

| Test cond | Signal name | Pin no. | Vil (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| F21 | DATA A7-IN | 30 | 0.0 | -10 | 10 | uA | 1-3 |
| F22 | DATA B0-IN | 33 | | | | | |
| F23 | DATA B1-IN | 38 | | | | | |
| F24 | DATA B2-IN | 41 | | | | | |
| F25 | DATA B3-IN | 44 | | | | | |
| F26 | DATA B4-IN | 48 | 0.0 | -10 | 10 | uA | 1-3 |

NOTES:

1. Input current, low level, test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 5.5 V | ± 55 mV | F1-F26 |
| Power supply current limit | 1.17 A | NA | F1-F26 |
| Forcing voltage, (Vil) | 0.0 V | ± 100 mV | F1-F26 |
| Pattern voltage high, (VH) | 3.0 V | ± 100 mV | F1-F26 |
| low, (VL) | 0.0 V | ± 100 mV | F1-F26 |



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NOTES: - Continued.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Input current measurement | NA | ± 8 uA | F1,F2 |
| | NA | ± 5 uA | F3,F4 |
| | NA | ± 13 uA | F5-F13 |
| | NA | ± 1 uA | F14-F26 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

(e) All floating inputs shall be tied to 3.0 V.

3. Abbreviations used are defined in 6.1.



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TABLE VIII. Output voltage, high level, test.

| Test cond | Signal name | Pin no. | Ioh (mA) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| G1 | DATA A0-OUT | 52 | -5.0 | 2.4 | 4.5 | V | 1-4 |
| G2 | DATA A1-OUT | 56 | | | | | |
| G3 | DATA A2-OUT | 59 | | | | | |
| G4 | DATA A3-OUT | 64 | | | | | |
| G5 | DATA A4-OUT | 68 | | | | | |
| G6 | DATA A5-OUT | 72 | | | | | |
| G7 | DATA A6-OUT | 76 | | | | | |
| G8 | DATA A7-OUT | 80 | | | | | |
| G9 | DATA B0-OUT | 83 | | | | | |
| G10 | DATA B1-OUT | 88 | | | | | |
| G11 | DATA B2-OUT | 91 | | | | | |
| G12 | DATA B3-OUT | 94 | | | | | |
| G13 | DATA B4-OUT | 98 | -5.0 | 2.4 | 4.5 | V | 1-4 |
| G14 | PU | 84 | -0.2 | 2.4 | 2.6 | V | 1-4 |



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NOTES:

1. Output voltage, high level, test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 4.5 V | ± 45 mV | G1-G14 |
| Power supply current limit | 1.17 A | NA | G1-G14 |
| Forcing current, (Ioh)<br>(Ioh) | -5.0 mA<br>-0.2 mA | ± 50 uA<br>± 2 uA | G1-G13<br>G14 |
| Pattern voltage high, (VH)<br>low, (VL) | 3.0 V<br>0.0 V | ± 100 mV<br>± 100 mV | G1-G14<br>G1-G14 |
| Output voltage measurement | NA<br>NA | ± 105 mV<br>± 10 mV | G1-G13<br>G14 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.



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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

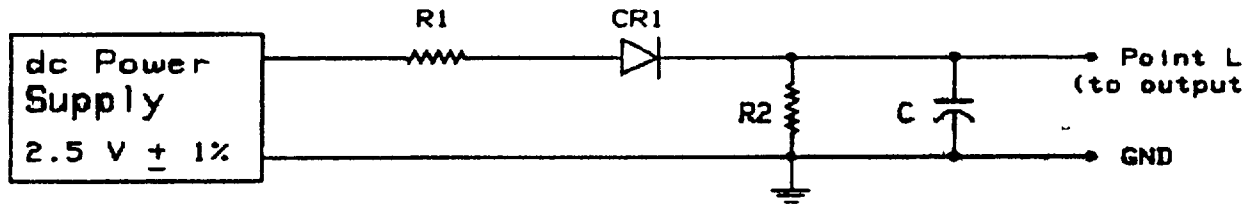
(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VIII and the voltage then measured.

3. The lead shall be as indicated on figure 6.

4. Abbreviations used are defined in 6.1.



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**NOTES:**

1. **CR1 = 1N4148 or equivalent.**

2. **The load capacitance is defined as follows:**

    (a) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

    (b) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

    (c) C = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk and disable time tests.

    (d) **The functional test shall be performed with only a 150** ± 15 pF load (including probe and parasitic capacitance) attached to each output.

    (e) C not required for any other tests.

FIGURE 6. Logic load circuit.



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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

| Pin number | R2 1/ (ohms) | Iol (mA) | Vol (volts) |
|---|---|---|---|
| 52,56,59,64,68, 72,76,80,83,88, 91,94,98 | 480 | 4.0 | 0.4 |
| 84 | 12000 | NA | NA |

1/ All resistors are metal film, ± 1 percent, 1/4 W.



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TABLE IX. Output voltage, low level, test.

| Test cond | Signal name | Pin no. | Iol (mA) | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| H1 | DATA A0-OUT | 52 | 4.0 | 0.0 | 0.4 | V | 1-4 |
| H2 | DATA A1-OUT | 56 | | | | | |
| H3 | DATA A2-OUT | 59 | | | | | |
| H4 | DATA A3-OUT | 64 | | | | | |
| H5 | DATA A4-OUT | 68 | | | | | |
| H6 | DATA A5-OUT | 72 | | | | | |
| H7 | DATA A6-OUT | 76 | | | | | |
| H8 | DATA A7-OUT | 80 | | | | | |
| H9 | DATA B0-OUT | 83 | | | | | |
| H10 | DATA B1-OUT | 88 | | | | | |
| H11 | DATA B2-OUT | 91 | | | | | |
| H12 | DATA B3-OUT | 94 | | | | | |
| H13 | DATA B4-OUT | 98 | 4.0 | 0.0 | 0.4 | V | 1-4 |



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NOTES:

1. Output voltage, low level, test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 4.5 V | ± 45 mV | H1-H13 |
| Power supply current limit | 1.17 A | NA | H1-H13 |
| Forcing current, (Iol) | 4.0 mA | ± 40 uA | H1-H13 |
| Pattern voltage high, (VH)<br>low, (VL) | 3.0 V<br>0.0 V | ± 100 mV<br>± 100 mV | H1-H13<br>H1-H13 |
| Output voltage measurement | NA | ± 20 mV | H1-H13 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.



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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 6.

4. Abbreviations used are defined in 6.1.



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TABLE X. Output leakage current, high level, test.

| Test cond | Signal name | Pin no. | Forcing voltage (V) | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| J1 | DATA A0-OUT | 52 | 5.5 | -10 | 10 | uA | 1-3 |
| J2 | DATA A1-OUT | 56 | | | | | |
| J3 | DATA A2-OUT | 59 | | | | | |
| J4 | DATA A3-OUT | 64 | | | | | |
| J5 | DATA A4-OUT | 68 | | | | | |
| J6 | DATA A5-OUT | 72 | | | | | |
| J7 | DATA A6-OUT | 76 | | | | | |
| J8 | DATA A7-OUT | 80 | | | | | |
| J9 | DATA B0-OUT | 83 | | | | | |
| J10 | DATA B1-OUT | 88 | | | | | |
| J11 | DATA B2-OUT | 91 | | | | | |
| J12 | DATA B3-OUT | 94 | | | | | |
| J13 | DATA B4-OUT | 98 | 5.5 | -10 | 10 | uA | 1-3 |



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NOTES:

1. Output leakage current, high level, test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 5.5 V | ± 55 mV | J1-J13 |
| Power supply current limit | 1.17 A | NA | J1-J13 |
| Forcing voltage, (Voh) | 5.5 V | ± 275 mV | J1-J13 |
| Pattern voltage high, (VH)<br>low, (VL) | 3.0 V<br>0.0 V | ± 100 mV<br>± 100 mV | J1-J13<br>J1-J13 |
| Output current measurement | NA | ± 1 uA | J1-J13 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.



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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table X and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.



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TABLE XI. Output leakage current, low level, test.

| Test cond | Signal name | Pin no. | Vol (V) | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| K1 | DATA A0-OUT | 52 | 0.0 | -10 | 10 | uA | 1-3 |
| K2 | DATA A1-OUT | 56 | | | | | |
| K3 | DATA A2-OUT | 59 | | | | | |
| K4 | DATA A3-OUT | 64 | | | | | |
| K5 | DATA A4-OUT | 68 | | | | | |
| K6 | DATA A5-OUT | 72 | | | | | |
| K7 | DATA A6-OUT | 76 | | | | | |
| K8 | DATA A7-OUT | 80 | | | | | |
| K9 | DATA B0-OUT | 83 | | | | | |
| K10 | DATA B1-OUT | 88 | | | | | |
| K11 | DATA B2-OUT | 91 | | | | | |
| K12 | DATA B3-OUT | 94 | | | | | |
| K13 | DATA B4-OUT | 98 | 0.0 | -10 | 10 | uA | 1-3 |



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NOTES:

1. Output leakage current, low level, test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 5.5 V | ± 55 mV | K1-K13 |
| Power supply current limit | 1.17 A | NA | K1-K13 |
| Forcing voltage, (Vol) | 0.0 V | ± 100 mV | K1-K13 |
| Pattern voltage high, (VH)<br>low, (VL) | 3.0 V<br>0.0 V | ± 100 mV<br>± 100 mV | K1-K13<br>K1-K13 |
| Output current measurement | NA | ± 1 uA | K1-K13 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.



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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.



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TABLE XII. Power supply current test.

| Test cond | Parameter | Pin no. | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|
| L1 | Icc (write) | 1,31 | 39.0 | 780.0 | mA | 1-3 |
| L2 | Icc (standby) | 1,31 | 6.5 | 130.0 | mA | 1-3 |

NOTES:

1. Power supply current test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 5.5 V | ± 55 mV | L1,L2 |
| Power supply current limit | 1.17 A | NA | L1,L2 |
| Pattern voltage<br>high, (VH)<br>low, (VL) | <br>3.0 V<br>0.0 V | <br>± 100 mV<br>± 100 mV | <br>L1,L2<br>L1,L2 |
| Power supply current measurement | <br>NA<br>NA | <br>± 37.0 mA<br>± 6.17 mA | <br>L1<br>L2 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.



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NOTES: - Continued.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XII.

3. Abbreviations used are defined in 6.1.



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TABLE XIII. Functional test.

| Test cond | Test description | Notes |
|---|---|---|
| M1 | Functional test<br>minimum power supply voltage<br>maximum pattern rate | 1-3 |
| M2 | Functional test<br>maximum power supply voltage<br>maximum pattern rate | 1-3 |

NOTES:

1. Functional test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 4.5 V<br>5.5 V | ± 45 mV<br>± 55 mV | M1<br>M2 |
| Power supply current limit | 1.17 A | NA | M1,M2 |
| Pattern voltage<br>high, (VH)<br>low, (VL) | <br>2.6 V<br>0.4 V | <br>± 130 mV<br>± 20 mV | <br>M1,M2<br>M1,M2 |
| Output voltage verification<br>high, (Voh)<br>high, (Voh)<br>low, (Vol) | <br>NA<br>NA<br>NA | <br>± 105 mV<br>± 155 mV<br>± 20 mV | <br>M1<br>M2<br>M1,M2 |



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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XIII using only a capacitive load of 150 pF (see figure 6). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.4\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 7.

3. Abbreviations used are defined in 6.1.



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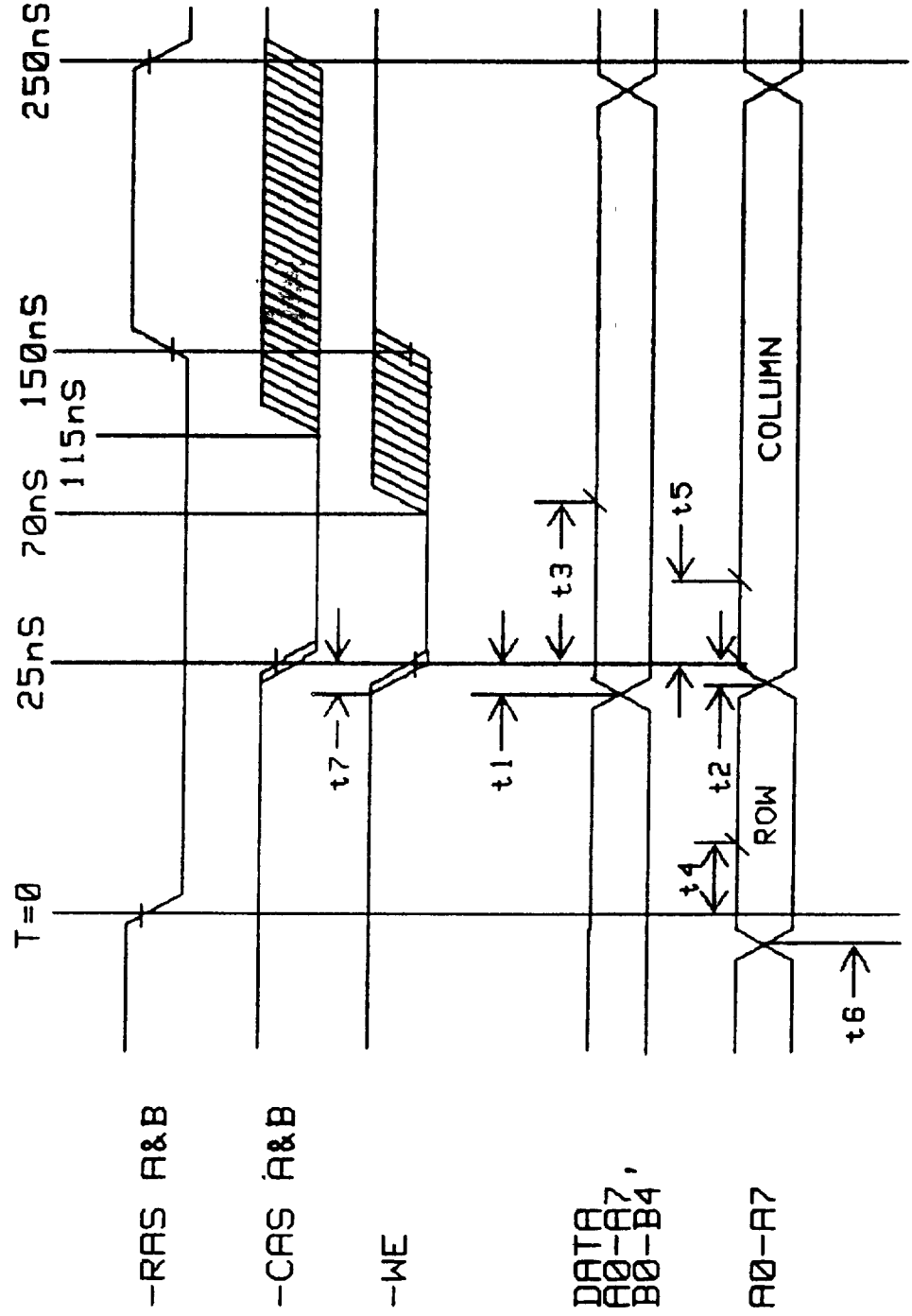



FIGURE 7. Input timing parameters for table XIII.



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NOTES:

1. Setup times:
   t1 = t2 = t6 = t7 = 0 nS minimum.

2. Hold times:
   t3 = 45 nS minimum.
   t4 = 20 nS minimum.
   t5 = 30 nS minimum.

3. The rise times and fall time are 3.5 ± 1.5 nS.



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TABLE XIV. Dynamic test.

| Test cond | Test description | Notes |
|---|---|---|
| N1 | Dynamic test, sample<br>maximum pattern rate<br>minimum power supply | 1-7 |
| N2 | Dynamic test, sample<br>maximum pattern rate<br>maximum power supply | 1-7 |
| N3 | Dynamic test, qualification<br>maximum pattern rate<br>minimum power supply | 1-7 |
| N4 | Dynamic test, qualification<br>maximum pattern rate<br>maximum power supply | 1-7 |

NOTES:

1. Dynamic test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 4.5 V<br>5.5 V | ± 45 mV<br>± 55 mV | N1,N3<br>N2,N4 |
| Power supply current limit | 1.17 A | NA | N1-N4 |
| Input voltage<br>high, (VH)<br>low, (VL) | <br>2.4 V<br>0.4 V | <br>± 120 mV<br>± 20 mV | <br>N1-N4<br>N1-N4 |



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NOTES: - Continued.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Output voltage verification | | | |
| high, (Voh) | NA | ± 105 mV | N1,N3 |
| high, (Voh) | NA | ± 155 mV | N2,N4 |
| low, (Vol) | NA | ± 20 mV | N1-N4 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XV.

3. The load circuit for this test is specified on figure 6.

4. All of the input waveforms shall meet the requirements shown on figure 8.

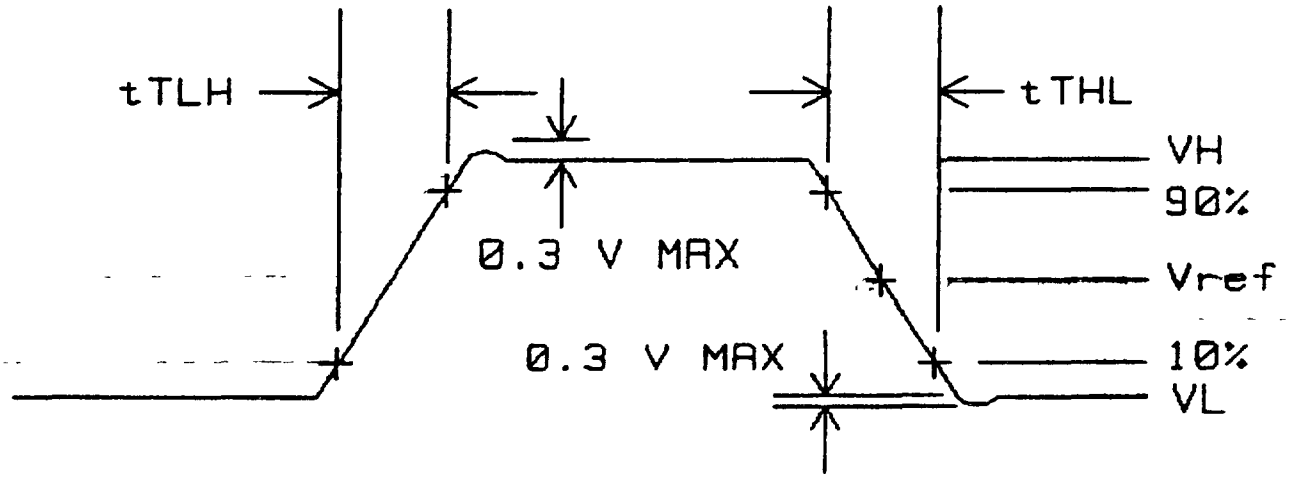
5. Input/Output (I/O) waveforms are as referenced on figure 9.

6. This test shall be performed as indicated in table XIV.

7. Abbreviations used are defined in 6.1.



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NOTES:

1. VH and VL are as specified in the test tolerance table.
2. Vref = 1.5 V.
3. Rise and fall times for the above waveform are as follows:

| Input pin | tTLH | tTHL |
|---|---|---|
| 2,3,4,6,7,8,9,11,14,18,22,26, 30,32,33,34,37,38,41,44,48,54, 57,58,82,87 | 3.5 ± 1.5 nS | 3.5 ± 1.5 nS |

FIGURE 8. Sample input waveform for table XIV.



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TABLE XV. Dynamic parameters.

| Ref num | Parameter number | Parameter | Minimum limit | | | Maximum limit | | | Units | Fig |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0°C | 25°C | 60°C | 0°C | 25°C | 60°C | | |
| 1 | 1-13 | Random read or write cycle time; tRC | 0.26 | 0.26 | 0.26 | 10.0 | 10.0 | 10.0 | uS | 9a,b,c |
| 2 | 14-26 | Read-write cycle time; tRMW | 0.28 | 0.28 | 0.28 | 10.0 | 10.0 | 10.0 | uS | 9d |
| 3 | 27-39 | Access time from -RAS A, -RAS B; tRAC | - | - | - | 156 | 156 | 156 | nS | 9a,d |
| 4 | 40-52 | Access time from -CAS A, -CAS B; tCAC | - | - | - | 96 | 96 | 96 | nS | 9a,d |
| 5 | 53-65 | Precharge time -RAS A, -RAS B; tRP | 100 | 100 | 100 | - | - | - | nS | 9a,b,c,d |
| 6 | 66-78 | Pulse width -RAS A, -RAS B; tRAS | 0.15 | 0.15 | 0.15 | 10 | 10 | 10 | uS | 9a,b,c,d |
| 7 | 79-91 | Hold time -RAS A, -RAS B; tRSH | 90 | 90 | 90 | - | - | - | nS | 9a,b,d |
| 8 | 92-104 | Hold time -CAS A, -CAS B; tCSH | 150 | 150 | 150 | - | - | - | nS | 9a,b,d |
| 9 | 105-117 | Pulse width -CAS A; -CAS B; tCAS | 0.09 | 0.09 | 0.09 | 10 | 10 | 10 | uS | 9a,b,d |
| 10 | 118-130 | Delay time -RAS A to -CAS A, -RAS B to -CAS B; tRCD | 30 | 30 | 30 | 60 | 60 | 60 | nS | 9a,b,d |
| 11 | 131-143 | Precharge time -CAS A to -RAS A, -CAS B to -RAS B; tCRP | 0 | 0 | 0 | - | - | - | nS | 9a,b,d |





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TABLE XV. Dynamic parameters - Continued.

| Ref num | Parameter number | Parameter | Minimum limit | | | Maximum limit | | | Units | Fig |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0°C | 25°C | 60°C | 0°C | 25°C | 60°C | | |
| 12 | 144-156 | Row address to -RAS A, -RAS B, setup time; tASR | 0 | 0 | 0 | - | - | - | nS | 9a,b,c,d |
| 13 | 157-169 | Column address to -CAS A, -CAS B setup time; tASC | 0 | 0 | 0 | - | - | - | nS | 9a,b,d |
| 14 | 170-182 | Row address to -RAS A, -RAS B hold time; tRAH | 20 | 20 | 20 | - | - | - | nS | 9a,b,c,d |
| 15 | 183-195 | Column address to -CAS A, -CAS B hold time; tCAH | 45 | 45 | 45 | - | - | - | nS | 9a,b,d |
| 16 | 196-208 | Column address to -CAS A, -CAS B hold time referenced to -RAS A, -RAS B; tAR | 100 | 100 | 100 | - | - | - | nS | 9a,b,d |
| 17 | 209-221 | Read command setup time; tRCS | 0 | 0 | 0 | - | - | - | nS | 9a,d |
| 18 | 222-234 | Read command hold time; tRCH | 0 | 0 | 0 | - | - | - | nS | 9a |
| 19 | 235-247 | Write command hold time; tWCH | 45 | 45 | 45 | - | - | - | nS | 9b |
| 20 | 248-260 | Write command hold time referenced to -RAS A, -RAS B; tWCR | 120 | 120 | 120 | - | - | - | nS | 9b |
| 21 | 261-273 | Write command pulse width; tWP | 45 | 45 | 45 | - | - | - | nS | 9b,d |



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TABLE XV. Dynamic parameters - Continued.

| Ref num | Parameter number | Parameter | Minimum limit | | | Maximum limit | | | Units | Fig |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0°C | 25°C | 60°C | 0°C | 25°C | 60°C | | |
| 22 | 274-286 | Write command lead time to -RAS A, -RAS B; tRWL | 50 | 50 | 50 | - | - | - | nS | 9b,d |
| 23 | 287-299 | Write command lead time to -CAS A, -CAS B; tCWL | 50 | 50 | 50 | - | - | - | nS | 9b,d |
| 24 | 300-312 | DATA-IN setup time; tDS | 0 | 0 | 0 | - | - | - | nS | 9b,d |
| 25 | 313-325 | DATA-IN hold time; tDH | 45 | 45 | 45 | - | - | - | nS | 9b,d |
| 26 | 326-338 | DATA-IN hold time referenced to -RAS A -RAS B; tDHR | 120 | 120 | 120 | - | - | - | nS | 9b |
| 27 | 339-351 | Refresh period; tREF | - | - | - | 4 | 4 | 4 | mS | 9b |
| 28 | 352-364 | Write command setup time; tWCS | 0 | 0 | 0 | - | - | - | nS | 9b |
| 29 | 365-377 | -CAS A, -CAS B to -WE delay; tCWD | 90 | 90 | 90 | - | - | - | nS | 9d |
| 30 | 378-390 | -RAS A, -RAS B to -WE delay; tRWD | 140 | 140 | 140 | - | - | - | nS | 9d |
| 31 | 391-403 | Read command hold time referenced to -RAS A, -RAS B; tRRH | 25 | 25 | 25 | - | - | - | nS | 9a |
| 3 | 404-416 | Access time from -RAS A, -RAS B; tRAC | - | - | - | 174 | 174 | 174 | nS | 9a,d |
| 4 | 417-429 | Access time from -CAS A, -CAS B; tCAC | - | - | - | 109 | 109 | 109 | nS | 9a,d |



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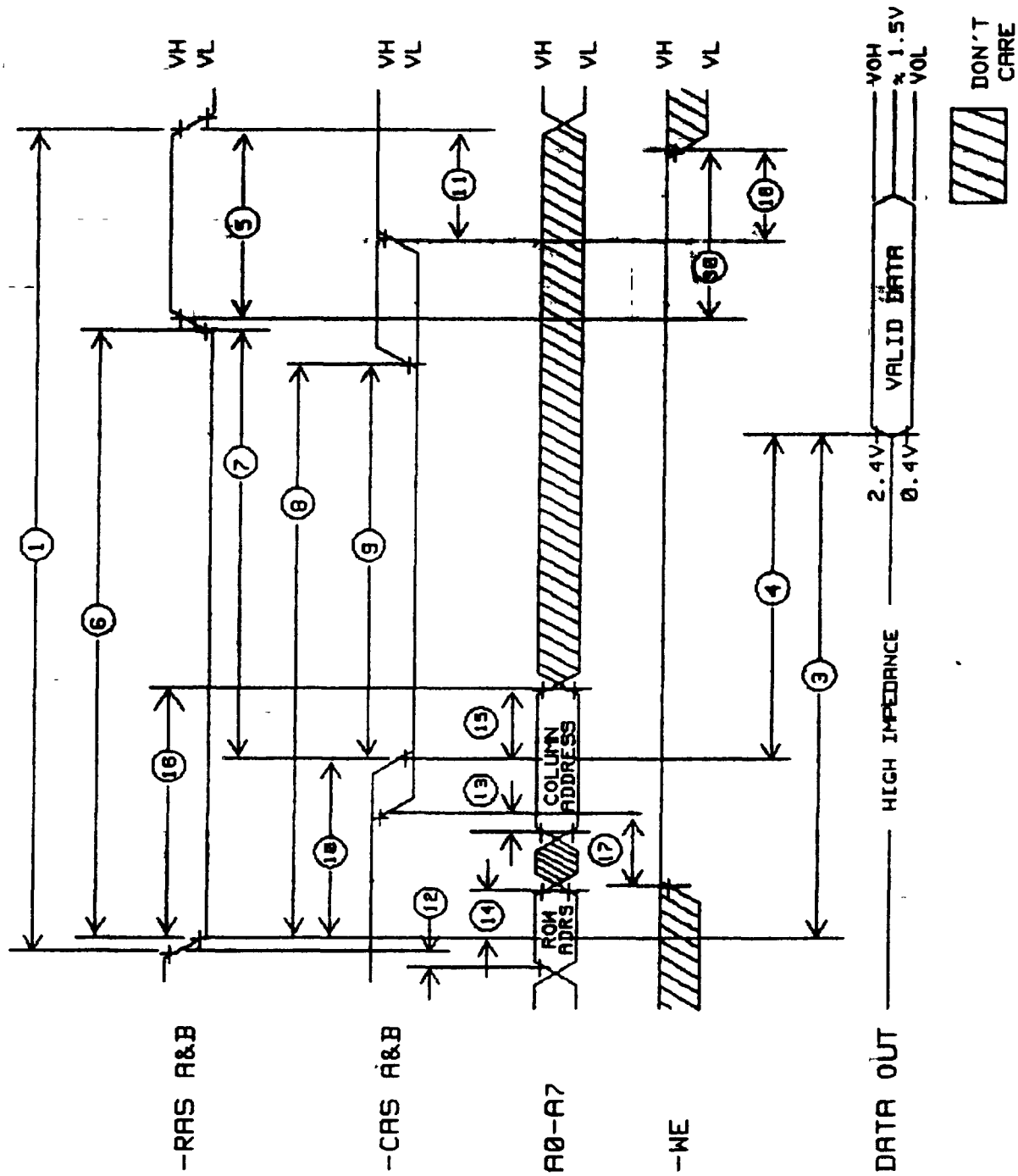



FIGURE 9a. Dynamic waveforms for table XV.



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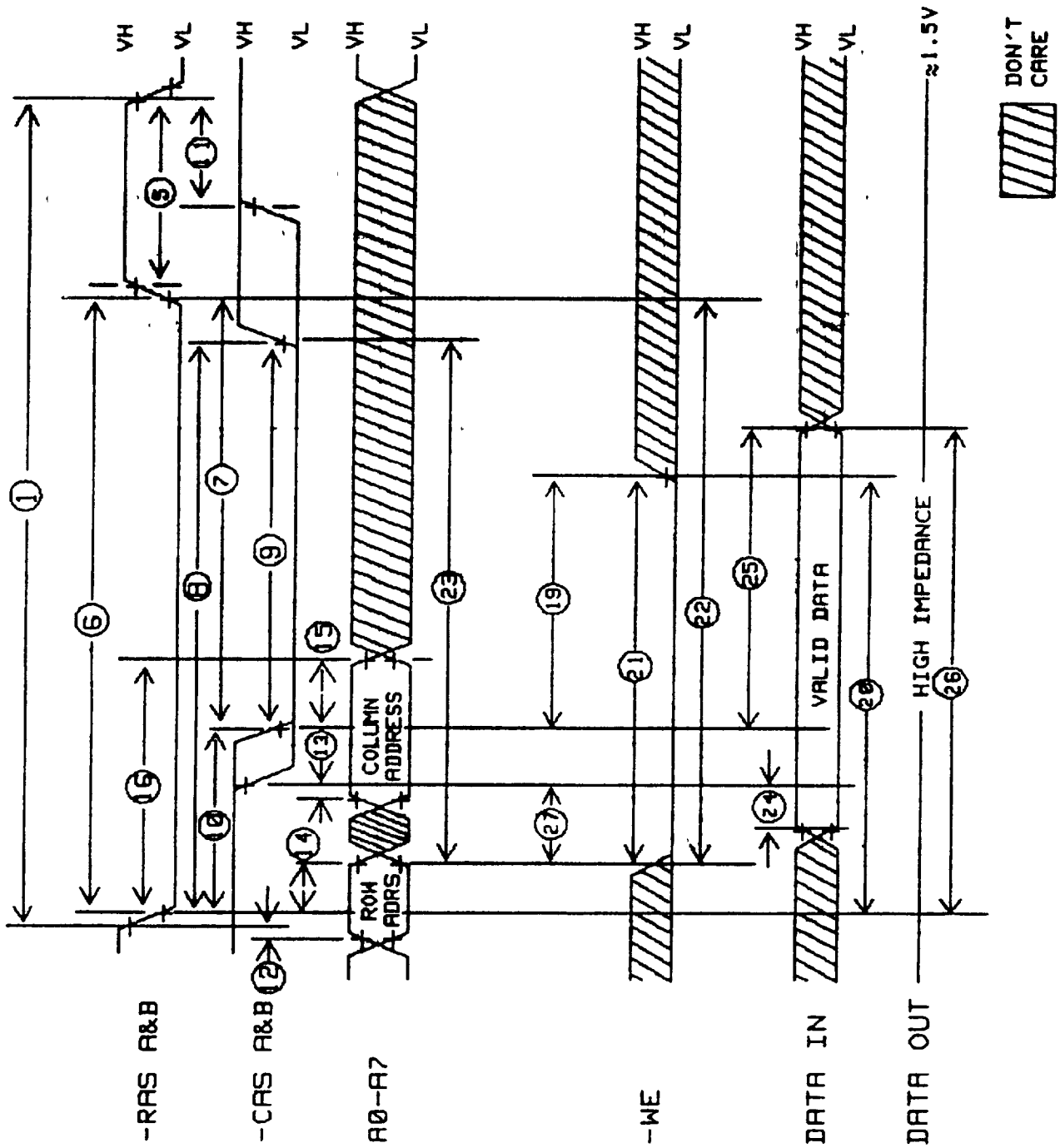



FIGURE 9b. Dynamic waveforms for table XV.



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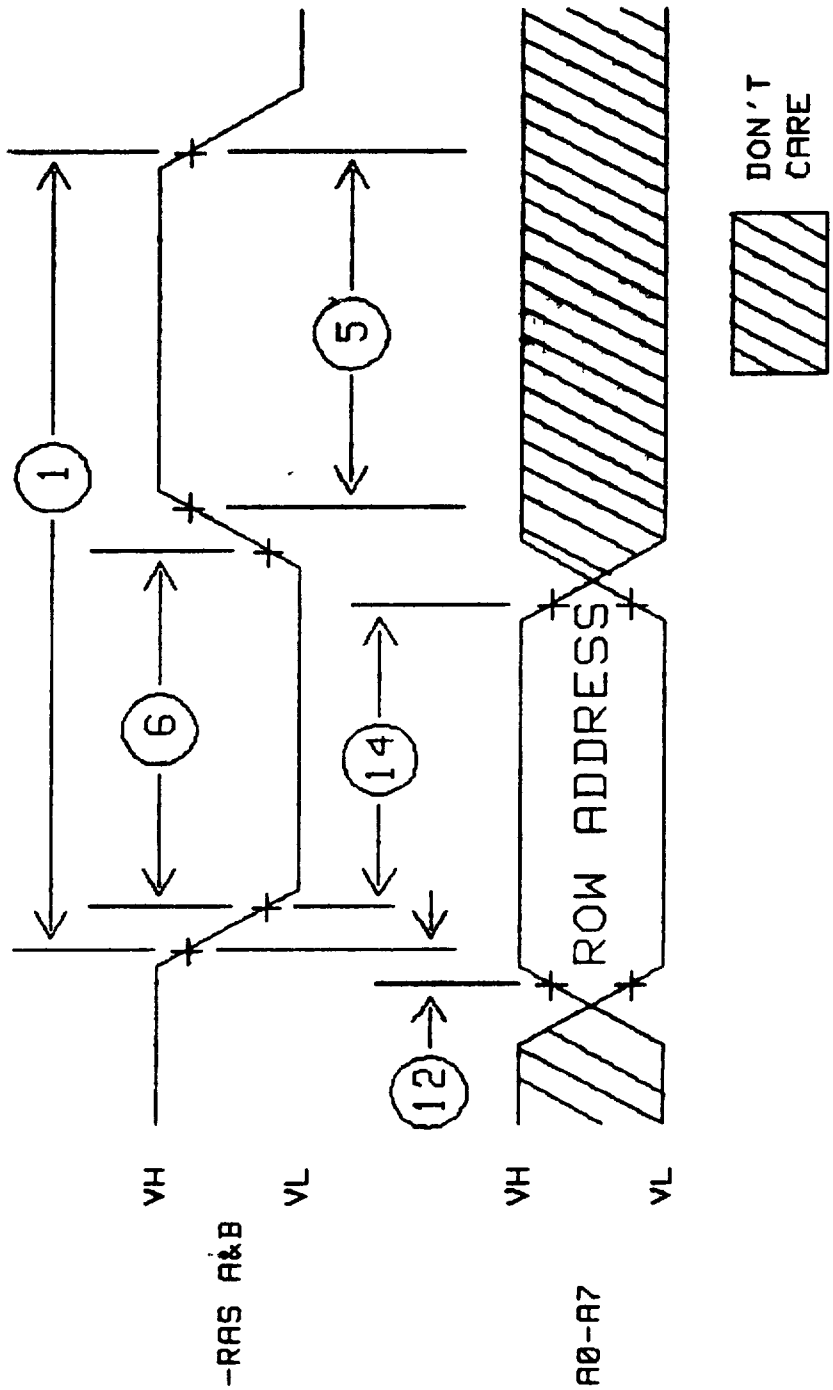
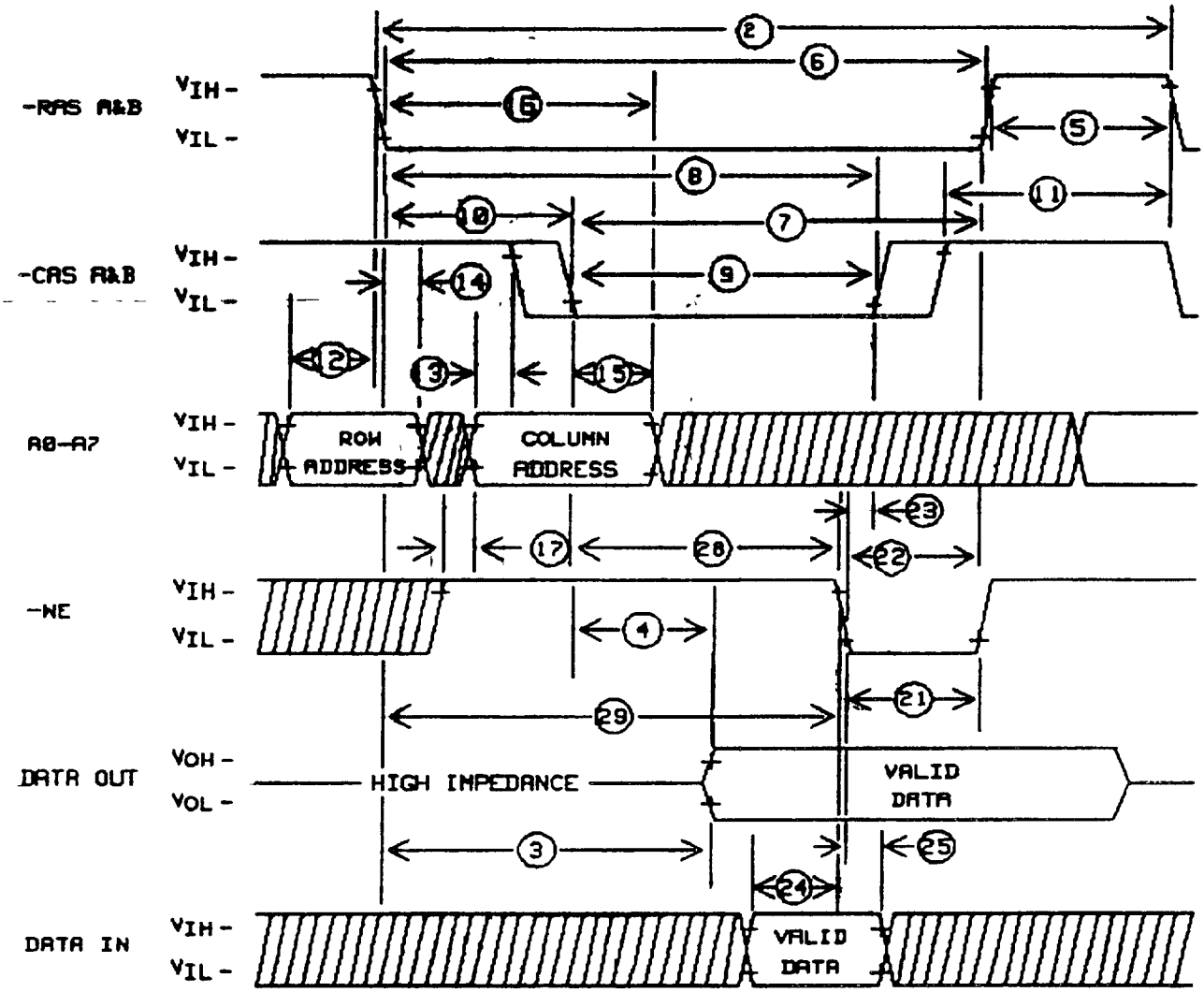



**FIGURE 9c.** Dynamic waveforms for table XV.



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NOTE:

1. Reference numbers 3 and 4 apply to test conditions N3 and N4 only.

FIGURE 9d. Dynamic waveforms for table XV.



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TABLE XVI. Power supply filter capacitance test.

| Test cond | Parameter | Pin numbers | 25 ± 5°C Initial limits Min | Max | Units | Note |
|---|---|---|---|---|---|---|
| P | PSFC | 1,31, 51,81 | 14.1 | 26.1 | uF | 1-3 |

NOTES:

1. Power supply filter capacitance test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Capacitance bridge frequency | 1 KHz | ± 100 Hz | P |
| Capacitance bridge voltage | 0.3 Vp | ± 30 mVp | P |
| Capacitance measurement | NA | ± 0.6 uF | P |

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground a the positive terminal to the pin indicated in table XVI.

3. Abbreviations used are defined in 6.1.



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TABLE XVII. Power supply transient amplitude test.

| Test cond | Parameter | Pin no. | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|
| Q | PSTA | 1,31 51,81 | 0 | 400 | mV | 1-4 |

NOTES:

1. Power supply transient amplitude test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 5.0 V | ± 250 mV | Q |
| Power supply current limit | 1.17 A | NA | Q |
| Pattern voltage high, (VH) low, (VL) | 3.0 V 0.0 V | ± 100 mV ± 100 mV | Q Q |
| Power supply transient amplitude measurement | NA | ± 20 mV | Q |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.



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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XVII.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to Vcc at the point of shorting.

(d) Using the oscilloscope, D07, inspect for the most negative spike as shown on figure 10 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 12 pF. There shall be no other capacitance present on Vcc to ground.

3. The module inputs shall not violate the minimum input timing parameters referenced on figure 7.

4. Abbreviations used are defined in 6.1.



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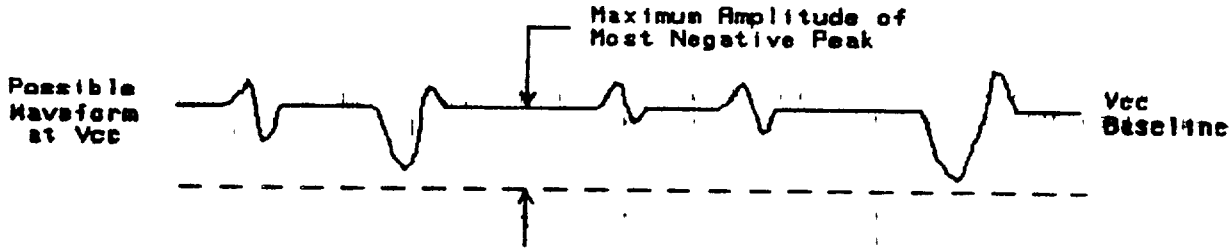



FIGURE 10. Power supply transient amplitude waveform for table XVII.



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TABLE XVIII. Input capacitance test.

| Test cond | Signal name | Pin no. | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|
| R1 | DATA A0-IN | 2 | 2 | 20 | pF | 1-3 |
| R2 | DATA A1-IN | 6 | | | | |
| R3 | DATA A2-IN | 9 | | | | |
| R4 | DATA A3-IN | 14 | | | | |
| R5 | DATA A4-IN | 18 | | | | |
| R6 | DATA A5-IN | 22 | | | | |
| R7 | DATA A6-IN | 26 | | | | |
| R8 | DATA A7-IN | 30 | | | | |
| R9 | DATA B0-IN | 33 | | | | |
| R10 | DATA B1-IN | 38 | | | | |
| R11 | DATA B2-IN | 41 | | | | |
| R12 | DATA B3-IN | 44 | | | | |
| R13 | DATA B4-IN | 48 | 2 | 20 | pF | 1-3 |
| R14 | A0 | 8 | 26 | 195 | pF | 1-3 |
| R15 | A1 | 58 | | | | |
| R16 | A2 | 11 | | | | |
| R17 | A3 | 4 | | | | |
| R18 | A4 | 3 | | | | |
| R19 | A5 | 54 | | | | |
| R20 | A6 | 57 | 26 | 195 | pF | 1-3 |



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TABLE XVIII. Input capacitance test - Continued.

| Test cond | Signal name | Pin no. | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|
| R21 | A7 | 7 | 26 | 195 | pF | 1-3 |
| R22 | -WE | 34 | 26 | 195 | pF | 1-3 |
| R23 | -RAS A | 87 | 16 | 120 | pF | 1-3 |
| R24 | -CAS A | 82 | 16 | 120 | pF | 1-3 |
| R25 | -RAS B | 37 | 10 | 75 | pF | 1-3 |
| R26 | -CAS B | 32 | 10 | 75 | pF | 1-3 |

NOTES:

1. Input capacitance test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Capacitance meter frequency | 1 MHz | ± 100 KHz | R1-R26 |
| Capacitance meter voltage | 15 mV | ± 1.5 mV | R1-R26 |
| Capacitance measurement | NA | ± 0.9 pF | R1-R13 |
| | NA | ± 8.45 pF | R14-R22 |
| | NA | ± 5.2 pF | R23,R24 |
| | NA | ± 3.25 pF | R25,R26 |

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. Abbreviations used are defined in 6.1.



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TABLE XIX. Output capacitance test.

| Test cond | Signal name | Pin no. | 25 ± 5°C Initial limits Min | Max | Units | Notes |
|---|---|---|---|---|---|---|
| S1 | DATA A0-OUT | 52 | 2 | 25 | pF | 1-3 |
| S2 | DATA A1-OUT | 56 | | | | |
| S3 | DATA A2-OUT | 59 | | | | |
| S4 | DATA A3-OUT | 64 | | | | |
| S5 | DATA A4-OUT | 68 | | | | |
| S6 | DATA A5-OUT | 72 | | | | |
| S7 | DATA A6-OUT | 76 | | | | |
| S8 | DATA A7-OUT | 80 | | | | |
| S9 | DATA B0-OUT | 83 | | | | |
| S10 | DATA B1-OUT | 88 | | | | |
| S11 | DATA B2-OUT | 91 | | | | |
| S12 | DATA B3-OUT | 94 | | | | |
| S13 | DATA B4-OUT | 98 | 2 | 25 | pF | 1-3 |



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NOTES:

1. Output capacitance test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Capacitance meter frequency | 1 MHz | ± 100 KHz | S1-S13 |
| Capacitance meter voltage | 15 mV | ± 1.5 mV | S1-S13 |
| Capacitance measurement | NA | ± 1.15 pF | S1-S13 |

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIX.

3. Abbreviations used are defined in 6.1.



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TABLE XX. Disable time test.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | 25 + 5°C Initial limits Min | 25 + 5°C Initial limits Max | Units | Figs |
|---|---|---|---|---|---|---|---|---|
| High to high impedance (tphz) | | | | | | | | |
| T1 | -CAS A to A0-OUT | 34,8,58, 11,4,3, 54,57,7 | Open | 52 | 0 | 46 | nS | 11, 12 |
| T2 | -CAS A to A1-OUT | | | 56 | | | | |
| T3 | -CAS A to A2-OUT | | | 59 | | | | |
| T4 | -CAS A to A3-OUT | | | 64 | | | | |
| T5 | -CAS A to A4-OUT | | | 68 | | | | |
| T6 | -CAS A to A5-OUT | | | 72 | | | | |
| T7 | -CAS A to A6-OUT | | | 76 | | | | |
| T8 | -CAS A to A7-OUT | | | 80 | | | | |
| T9 | -CAS B to B0-OUT | | | 83 | | | | |
| T10 | -CAS B to B1-OUT | 34,8,58, 11,4,3, 54,57,7 | Open | 88 | 0 | 46 | nS | 11, 12 |





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TABLE XX. Disable time test - Continued.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | 25 + 5°C Initial limits Min | 25 + 5°C Initial limits Max | Units | Figs |
|---|---|---|---|---|---|---|---|---|
| T11 | -CAS B to B2-OUT | 34,8,58, 11,4,3, 54,57,7 | Open | 91 | 0 | 46 | nS | 11,12 |
| T12 | -CAS B to B3-OUT | | | 94 | | | | |
| T13 | -CAS B to B4-OUT | 34,8,58, 11,4,3, 54,57,7 | Open | 98 | 0 | 46 | nS | 11,12 |
| Low to high impedance (tplz) | | | | | | | | |
| T14 | -CAS A to A0-OUT | 34 | 8,58,11, 4,3,54, 57,7 | 52 | 0 | 65 | nS | 11,12 |
| T15 | -CAS A to A1-OUT | | | 56 | | | | |
| T16 | -CAS A to A2-OUT | | | 59 | | | | |
| T17 | -CAS A to A3-OUT | | | 64 | | | | |
| T18 | -CAS A to A4-OUT | | | 68 | | | | |
| T19 | -CAS A to A5-OUT | 34 | 8,58,11, 4,3,54, 57,7 | 72 | 0 | 65 | nS | 11,12 |





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TABLE XX. Disable time test - Continued.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Figs |
|---|---|---|---|---|---|---|---|---|
| T20 | -CAS A to A6-OUT | 34 | 8,58,11, 4,3,54, 57,7 | 76 | 0 | 65 | nS | 11, 12 |
| T21 | -CAS A to A7-OUT | | | 80 | | | | |
| T22 | -CAS B to B0-OUT | | | 83 | | | | |
| T23 | -CAS B to B1-OUT | | | 88 | | | | |
| T24 | -CAS B to B2-OUT | | | 91 | | | | |
| T25 | -CAS B to B3-OUT | | | 94 | | | | |
| T26 | -CAS B to B4-OUT | 34 | 8,58,11, 4,3,54, 57,7 | 98 | 0 | 65 | nS | 11, 12 |



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NOTES:

1. Disable time test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, (Vcc) | 5.0 V | ± 50 mV | T1-T26 |
| Power supply current limit | 1.17 A | NA | T1-T26 |
| Pattern voltage<br>high, (VH)<br>low, (VL) | <br>3.0 V<br>0.0 V | <br>± 100 mV<br>± 100 mV | <br>T1-T26<br>T1-T26 |
| Output time measurement | <br>NA<br>NA | <br>± 2 nS<br>± 3 nS | <br>T1-T13<br>T14-T26 |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

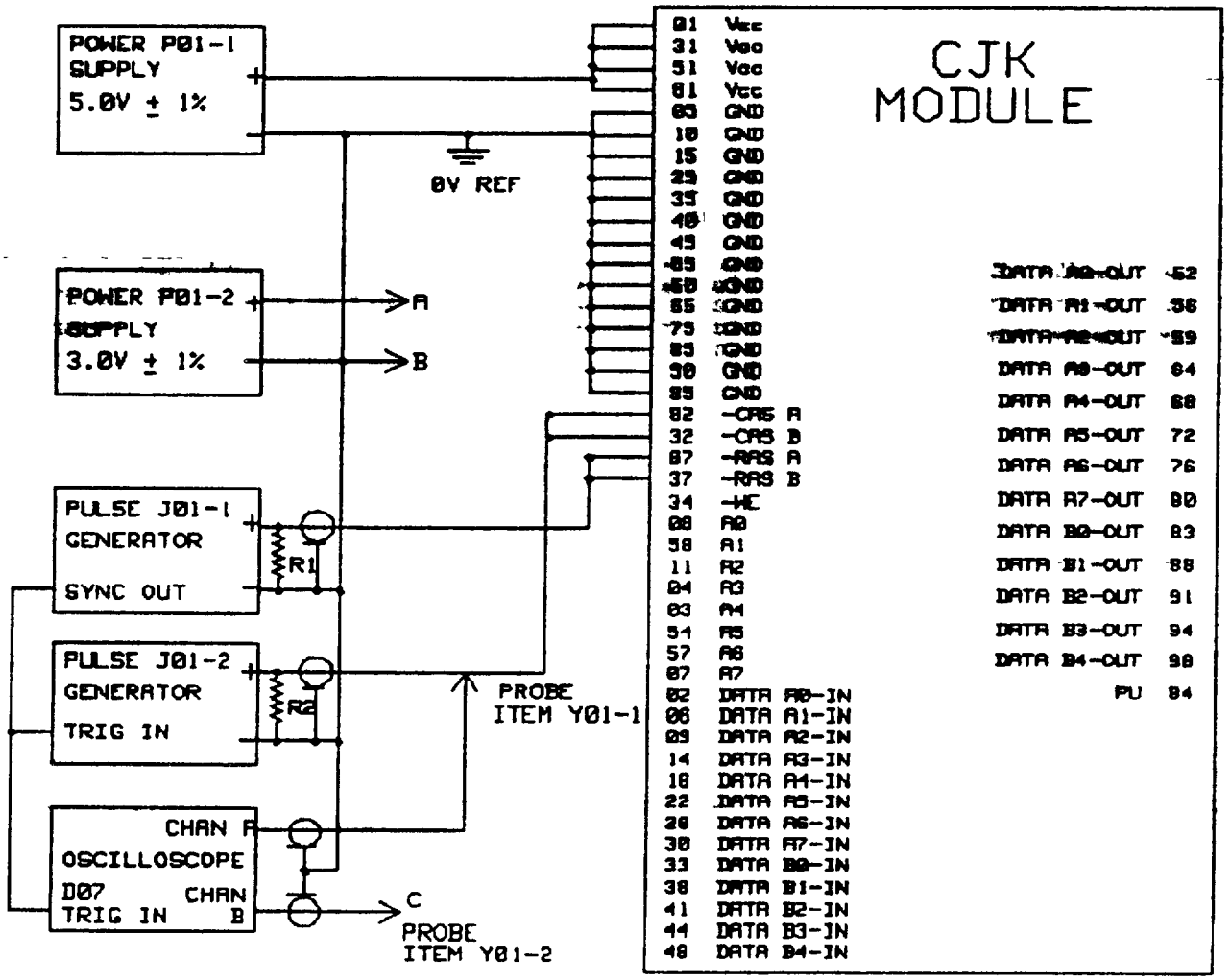
3. The test circuit shall be as indicated on figure 11.

4. Module inputs shall not violate the minimum input timing requirements referenced on figure 12.

5. Abbreviations used are defined in paragraph 6.1.



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NOTES:

1. R1 = R2 = 51 Ohm ± 5 percent, 1/2 W.

2. Output pins 52, 56, 59, 64, 68, 72, 76, 80, 83, 88, 91, 94 and 98 shall be loaded as specified on figure 6.

FIGURE 11. Disable time test circuit for table XX.



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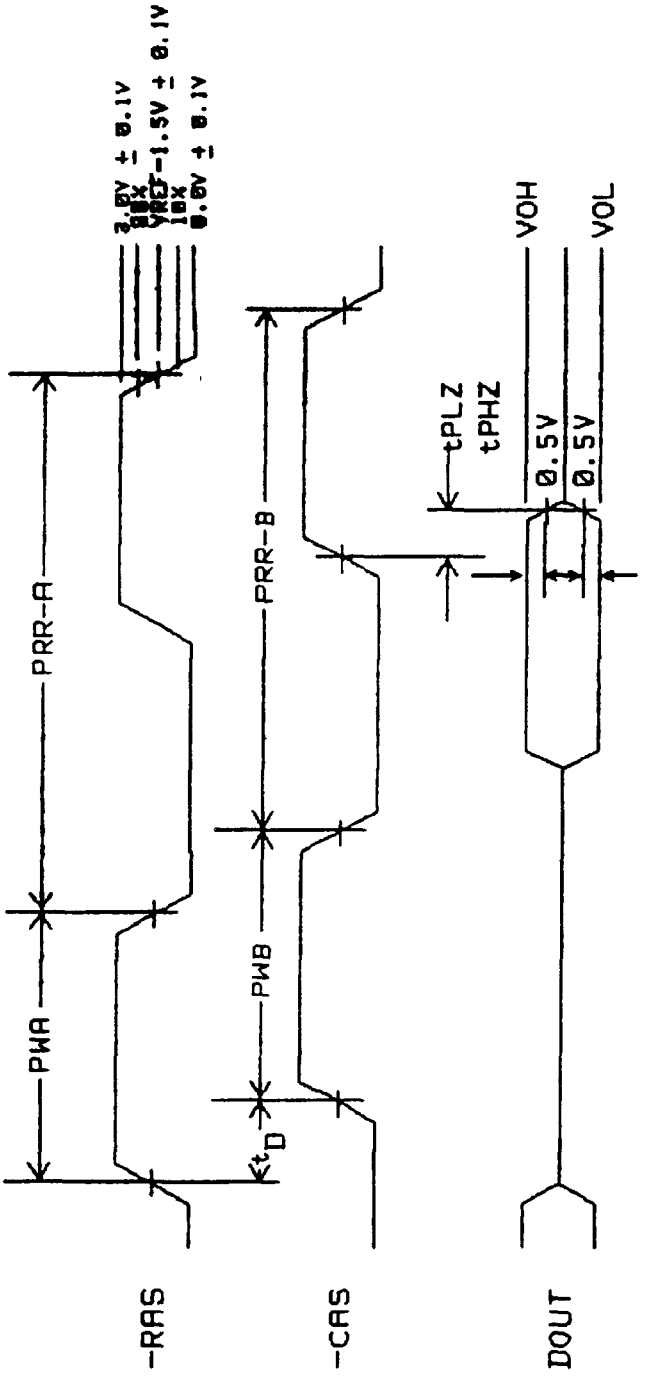



FIGURE 12. Disable time test waveforms for table XX.



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NOTES:

1. PWA = PWB = 250 nS ± 25 nS.
2. PRR-A = PRR-B = 500 nS ± 50 nS.
3. tD = 40 nS ± 5 nS.
4. Transition times shall be ≤ 10 nS.



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TABLE XXI. Threshold test.

| Test condition | Test description | Notes | Figure |
|---|---|---|---|
| U | Threshold test | 1-3 | 6 |

NOTES:

1. Threshold test tolerance table.

| Condition | Value | Accuracy | Test condition |
|---|---|---|---|
| Power supply, Vcc | 4.5 V | ± 45 mV | U |
| Power supply current limit | 1.17 A | NA | U |
| Pattern voltage<br>high, (VH)<br>low, (VL) | <br>2.4 V<br>0.8 V | <br>± 24 mV<br>± 8 mV | <br>U<br>U |
| Output voltage measurement<br>high, (Voh)<br>low, (Vol) | <br>NA<br>NA | <br>± 105 mV<br>± 20 mV | <br>U<br>U |

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table. No load is required for this test.



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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. Abbreviations used are defined in 6.1.



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[redacted]

6. PREPARING ACTIVITY

| a. NAME Technical Point of Contact (TPOC):<br>Mr. Robert F. Dilger (Code 602A)<br>PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS: | b. TELEPHONE (Include Area Code)<br>(1) Commercial: TPOC: 812-854-1528<br>(2) AUTOVON: 88-482-1528 |
|---|---|
| c. ADDRESS (Include Zip Code) Commanding Officer<br>Naval Weapons Support Center<br>ATTN: Robert F. Dilger, Bldg. 2940W, Code 602A<br>Crane, Indiana 47522-5060 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |

DD Form 1426, OCT 89 — Previous editions are obsolete. — 198/290

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